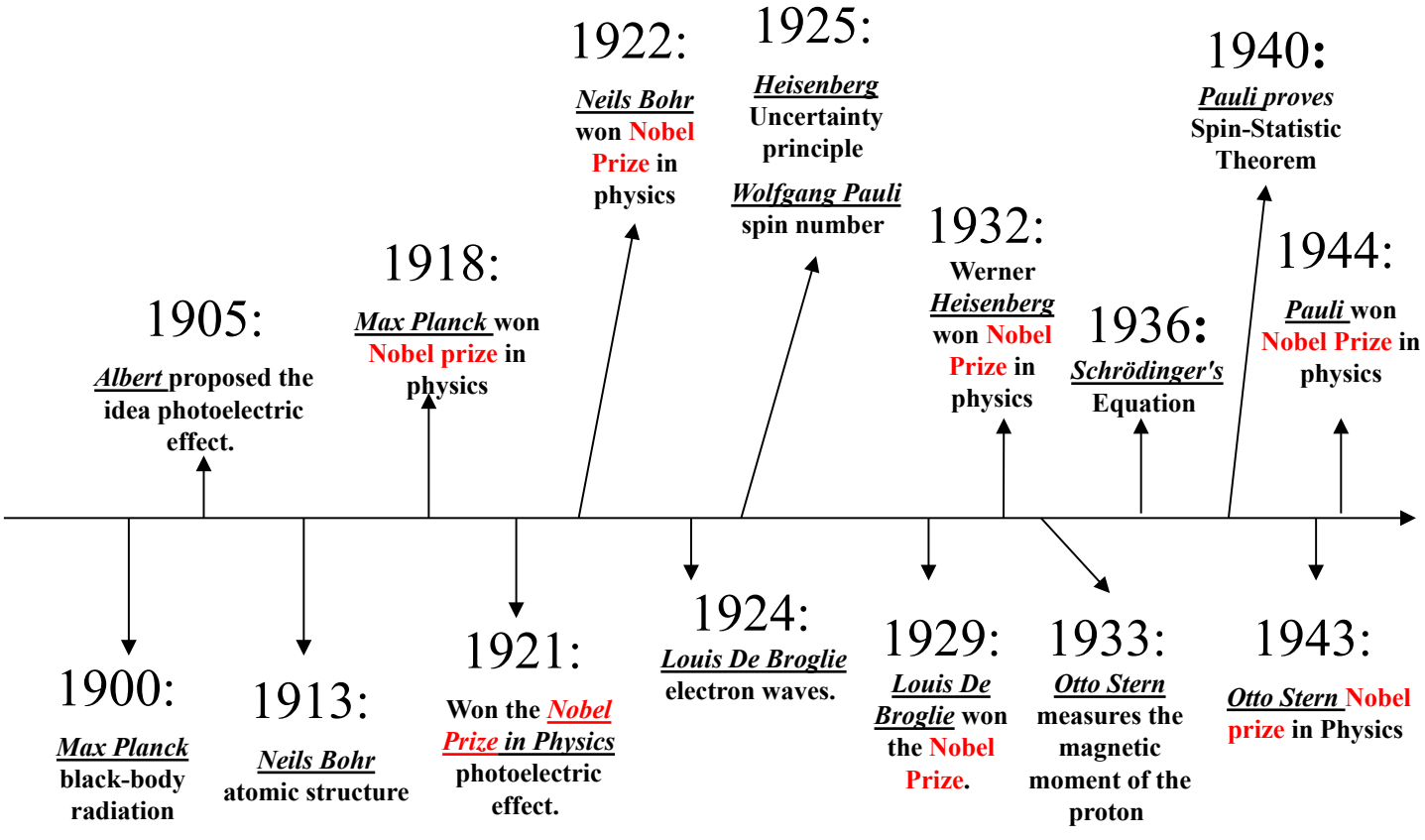
Quantum Chemistry

Dr. Saurav Prasad

Outline

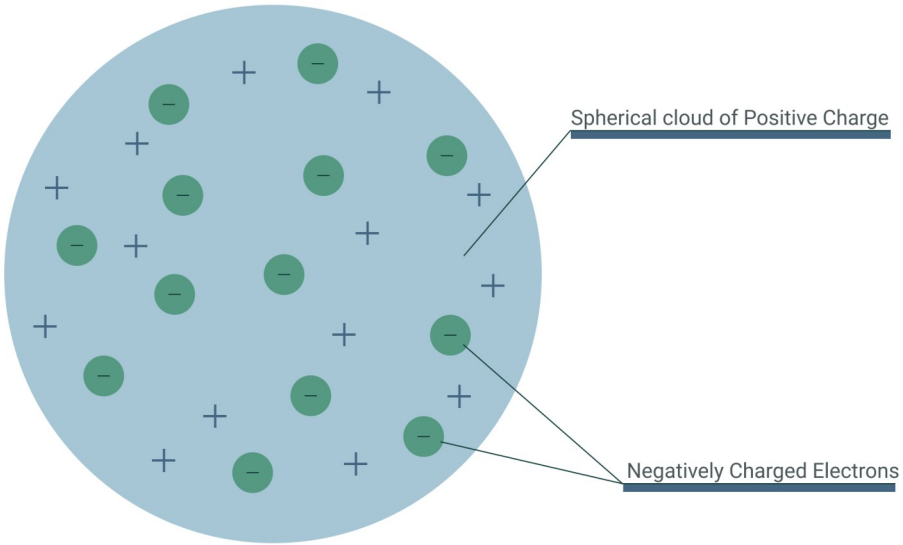
- Historical development,
- Bohr’s atomic model,
- de Broglie wavelength,
- Heisenberg uncertainty principle,
- Schrodinger equation,
- Wave function,
- particle in a box,
- Hydrogen atom,
- radial and angular solution to hydrogen atom,
- applications and limitations

Evolution of Quantum mechanics



Structure of the Atom

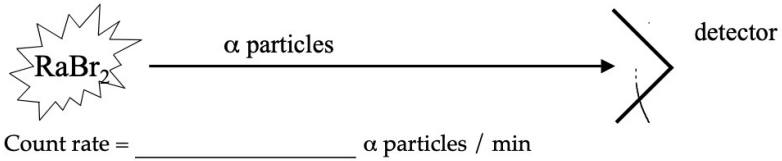
- In 1909, the prevailing theory of atomic structure was the plum pudding model.
- In this model, proposed by J. J. Thomson, the electrons were thought to be floating around in a soup of positive charge.

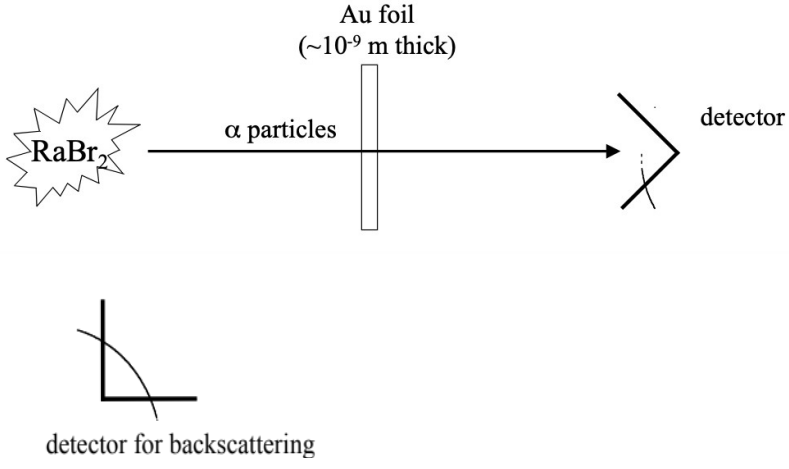


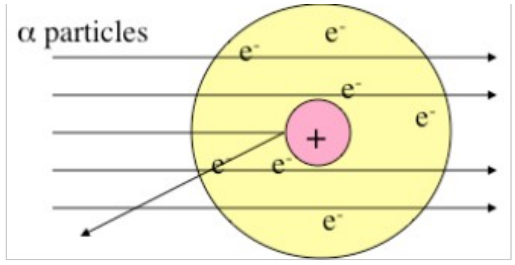
Rutherford's Atomic Model

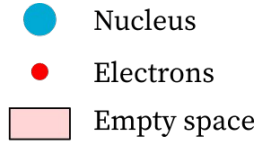
Ernest Rutherford (Cambridge University, England, 1871-1937) studied α emission from newly-discovered radioactive elements.

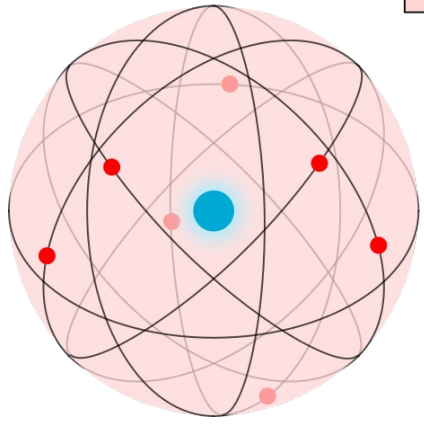
"It was almost as incredible as if you fired a fifteen-inch shell at a piece of tissue paper and it came back and hit you"









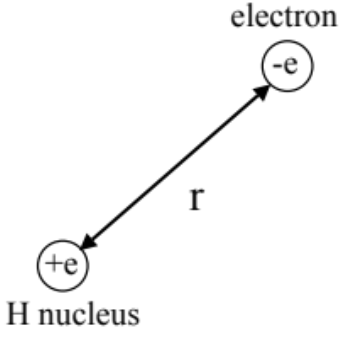


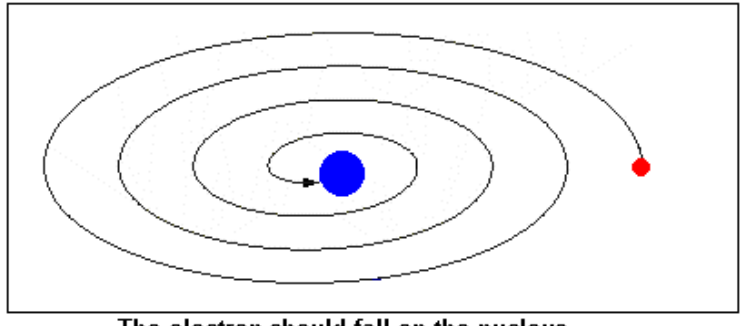
Rutherford's Atomic Model

Drawback

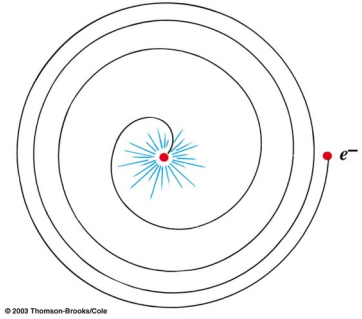
- According to classical theory, atoms in the model for Rutherford's are not stable.
- The motion of moving electrons should cause them to radiate energy in their orbits and to quickly execute a death spiral and collapse into the nucleus.
- The time taken for this collapse is estimated around 10^{-8} s, which is very small. According to the model, every atom will crumple in 10^{-8} s, but we know this is not reality.
- If the electrons are not revolving but stationary, still the electrons will fall into the nucleus by the electrostatic force between both.

The predicted death spiral of the electron!





The electron should fall on the nucleus.



Questions

Q1. Calculate the number of protons, neutrons and electrons in $^{80}_{35}Br$.

Q2. The number of electrons, protons and neutrons in a species are equal to 18, 16 and 16 respectively. Assign the proper symbol to the species.

Development leading to the Bohr's Model

- **Two developments played a major role in the formulation of Bohr's model of atom. These were**

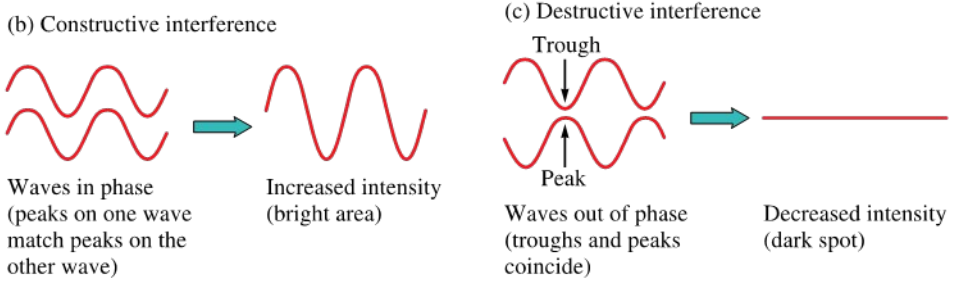
(i) Dual character of the electromagnetic radiation which means that radiations possess both wave like and particle like properties, and

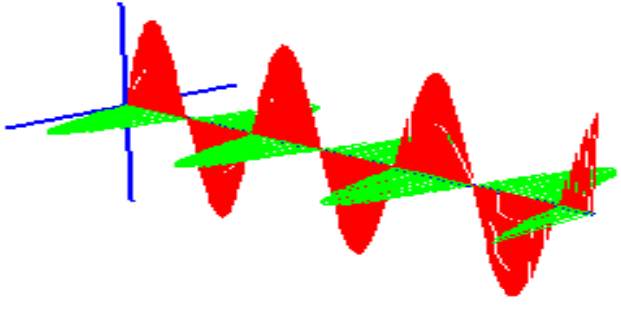
(ii) Experimental results regarding atomic spectra which can be explained only by assuming quantized electronic energy levels in atoms.

Wave nature of electromagnetic radiation

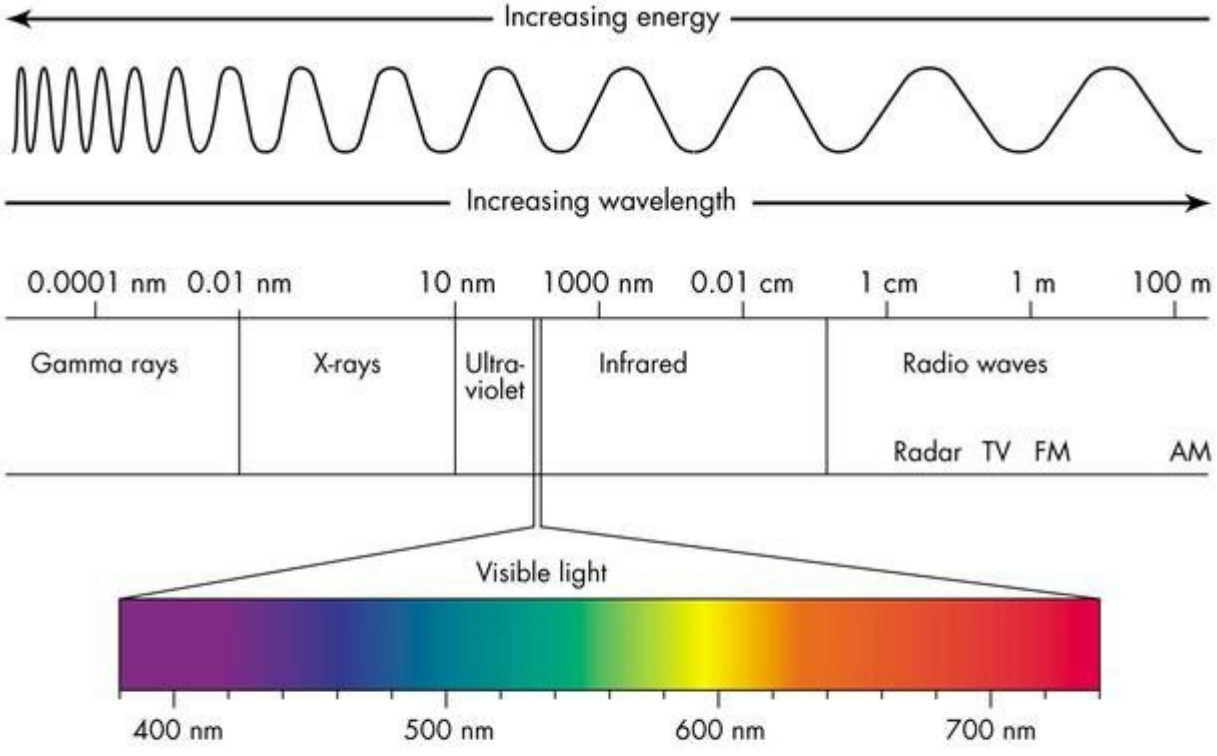
- John Maxwell, 1870: explained about the interaction between the charged bodies and the behaviour of electrical and magnetic fields on macroscopic level.

- He suggested that when electrically charged particle moves under acceleration, alternating electrical and magnetic fields are produced and transmitted. These fields are transmitted in the forms of waves called electromagnetic waves or electromagnetic radiation.





Electromagnetic Spectrum



$$c = \nu \lambda$$

Questions

Q1. The Vividh Bharati station of All India Radio, Delhi, broadcasts on a frequency of 1,368 kHz (kilo hertz). Calculate the wavelength of the electromagnetic radiation emitted by transmitter. Which part of the electromagnetic spectrum does it belong to?

Q2. The wavelength range of the visible spectrum extends from violet (400 nm) to red (750 nm). Express these wavelengths in frequencies (Hz).

Q3. Calculate (a) wavenumber and (b) frequency of yellow radiation having wavelength 5800 Å.

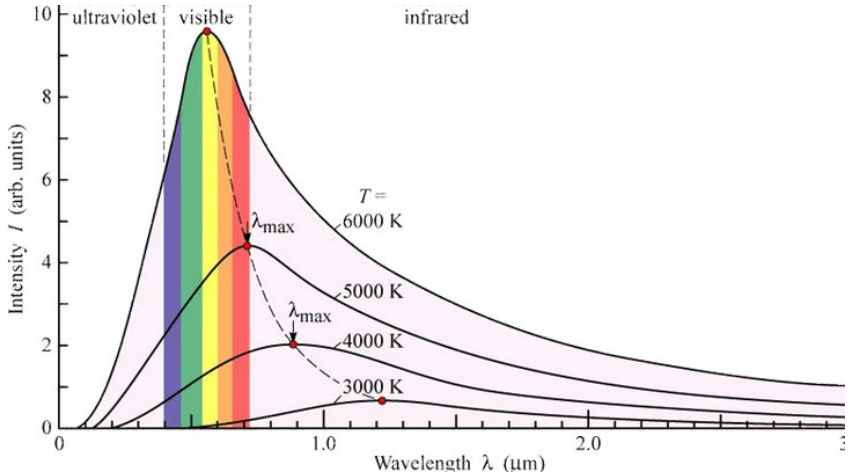
Particle Nature of Electromagnetic Radiation: Planck's QuantumTheory

- Diffraction and Interference: explained by wave nature of the electromagnetic radiation.
- Classically, particles and waves are distinct: Particles – characterised by position, mass, velocity; Waves – characterised by wavelength, frequency.
- Not explained by electromagnetic theory; 19th century Physics; Classical Physics:

1. the nature of emission of radiation from hot bodies (black -body radiation)
2. ejection of electrons from metal surface when radiation strikes it (photoelectric effect)
3. variation of heat capacity of solids as a function of temperature
4. line spectra of atoms with special reference to hydrogen.

Black Body Radiation

- The ideal body, which emits and absorbs radiations of all frequencies, is called a black body and the radiation emitted by such a body is called black body radiation.
- The exact frequency distribution of the emitted radiation (i.e., intensity versus frequency curve of the radiation) from a black body depends only on its temperature.



Black Body Radiation

- cannot be explained satisfactorily on the basis of the wave theory of light.
- Planck suggested that atoms and molecules could emit (or absorb) energy only in discrete quantities and not in a continuous manner.
- Planck gave the name quantum to the smallest quantity of energy that can be emitted or absorbed in the form of electromagnetic radiation.
- Energy of a quantum of photon

$$E = h\nu$$

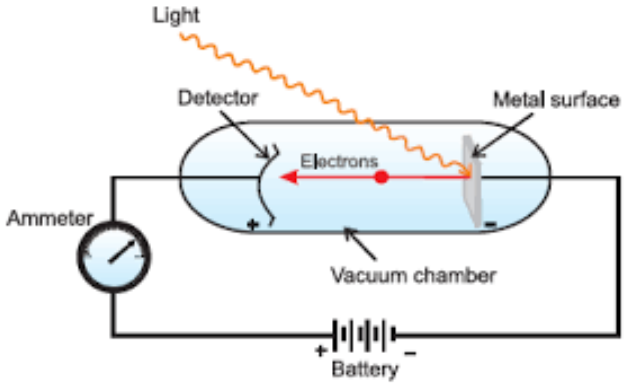
- h = Planck's constant = ??

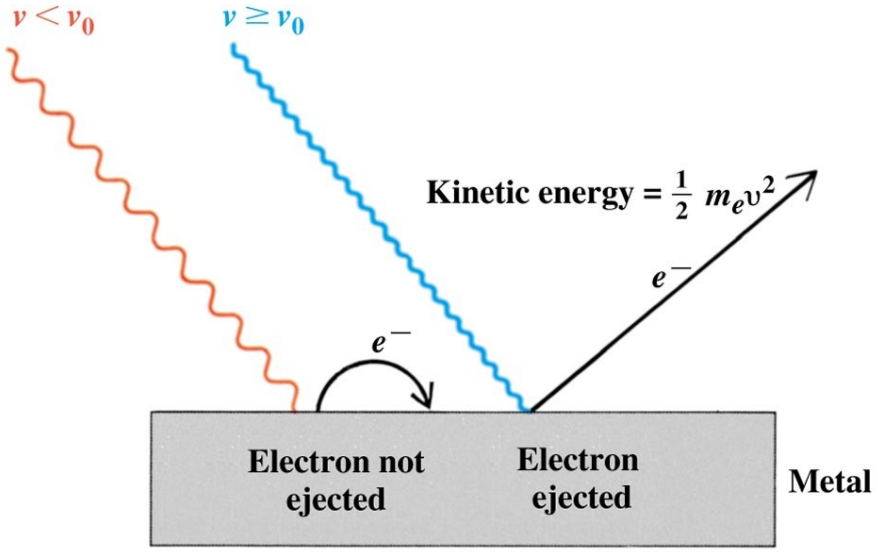
The Photoelectric Effect

A beam of light hitting a metal surface can cause electrons to be ejected from the surface.

Classical Paradigm: the energy of the ejected electrons should be proportional to the intensity (I) of the light and independent of the frequency (□) of the light.

Experiment: the energy of the ejected electrons is independent of the intensity (I) and depends directly on the frequency (□) of the light.





The Photoelectric Effect

The following observations characterize the photoelectric effect.

1.For light with frequency lower than the threshold frequency, no electrons are emitted regardless of the intensity of the light.

2. For light with frequency greater than the threshold frequency, the number of electrons emitted increases with the intensity of the light.

3. For light with frequency greater than the threshold frequency, the kinetic energy of the emitted electrons increases linearly with the frequency of the light.

The Photoelectric Effect

$$h\nu = h\nu_0 + \frac{1}{2} m_e v^2$$

Dual behaviour of electromagnetic radiation

- Particle nature: Explain black body radiation and photoelectric effect
- Wave nature: Diffraction and interference
- Light has dual character: microscopic particles like

Questions

Q1. Calculate energy of one mole of photons of radiation whose frequency is 5×10^{14} Hz.

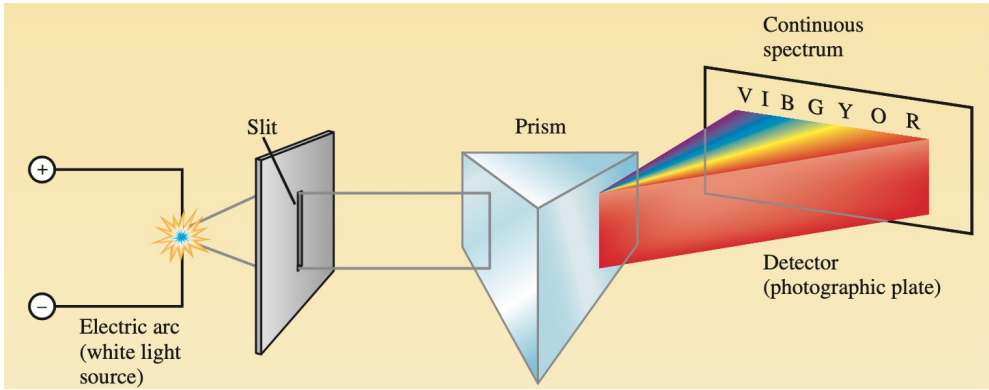
Q2. A 100 watt bulb emits monochromatic light of wavelength 400 nm. Calculate the number of photons emitted per second by the bulb.

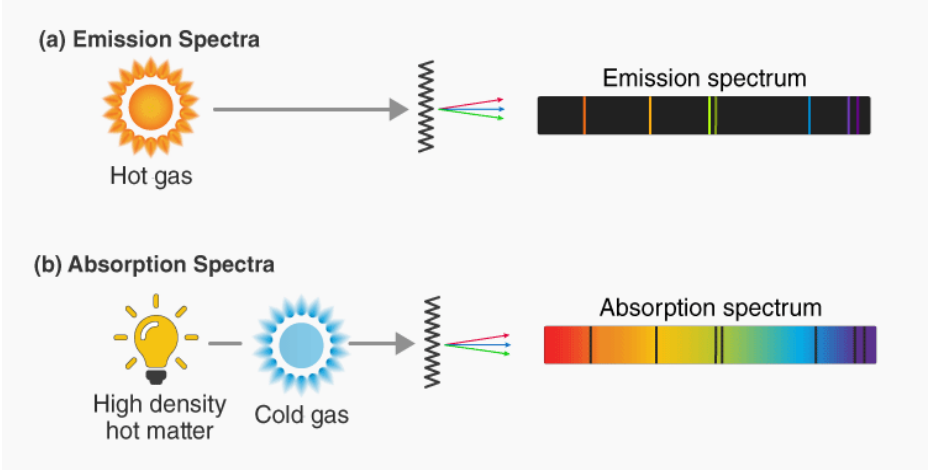
Q3. When electromagnetic radiation of wavelength 300 nm falls on the surface of sodium, electrons are emitted with a kinetic energy of 1.68×10^5 J mol^{-1}. What is the minimum energy needed to remove an electron from sodium? What is the maximum wavelength that will cause a photoelectron to be emitted ?

Q4. The threshold frequency v_0 for a metal is 7.0×10^{14} s^{-1}. Calculate the kinetic energy of an electron emitted when radiation of frequency $v = 1.0 \times 10^{15}$ s^{-1} hits the metal.

Hydrogen emission spectrum

The classical theory predicts that electromagnetic radiation is continuous



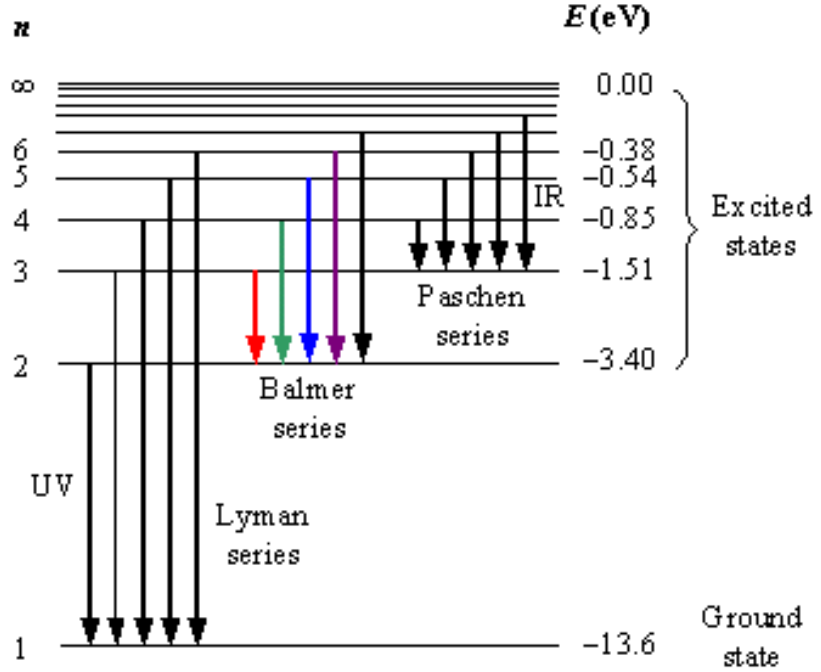


- When a high-energy discharge is passed through a sample of hydrogen gas, the H_2 molecules absorb energy, causing some of the H-H bonds to break.
- The resulting hydrogen atoms are *excited;* that is, they contain excess energy, which they release by emitting light of various wavelengths to produce what is called the *emission spectrum* of the hydrogen atom., called a **line spectrum**.

Hydrogen emission spectrum

- Each element has a unique line spectra.
- The characteristic lines in atomic spectra can be used in chemical analysis to identify unknown atoms in the same way as finger prints are used to identify people.

$$\bar{\nu} = 109{,}677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \ \text{cm}^{-1}$$



Energy levels of the hydrogen atom with some of the transitions between them that give rise to the spectral lines indicated.

Question

Q1. What are the frequency and wavelength of a photon emitted during a transition from n = 5 state to the n = 2 state in the hydrogen atom?

Some more questions!

Q1.(a) Calculate the KE of a photo electron emitted by sodium surface when light of wavelength 400 nm is incident on it. The work function of sodium is 3.653×10^{-19} J.

(b) Calculate the value of the longest wavelength which can result in the emission of a photoelectron from sodium surface.

Q2. A photon of wavelength 4×10^{-7} m strikes on metal surface, the work function of the metal being 2.13 eV. Calculate (i) the energy of the photon (eV), (ii) the kinetic energy of the emission, and (iii) the velocity of the photoelectron

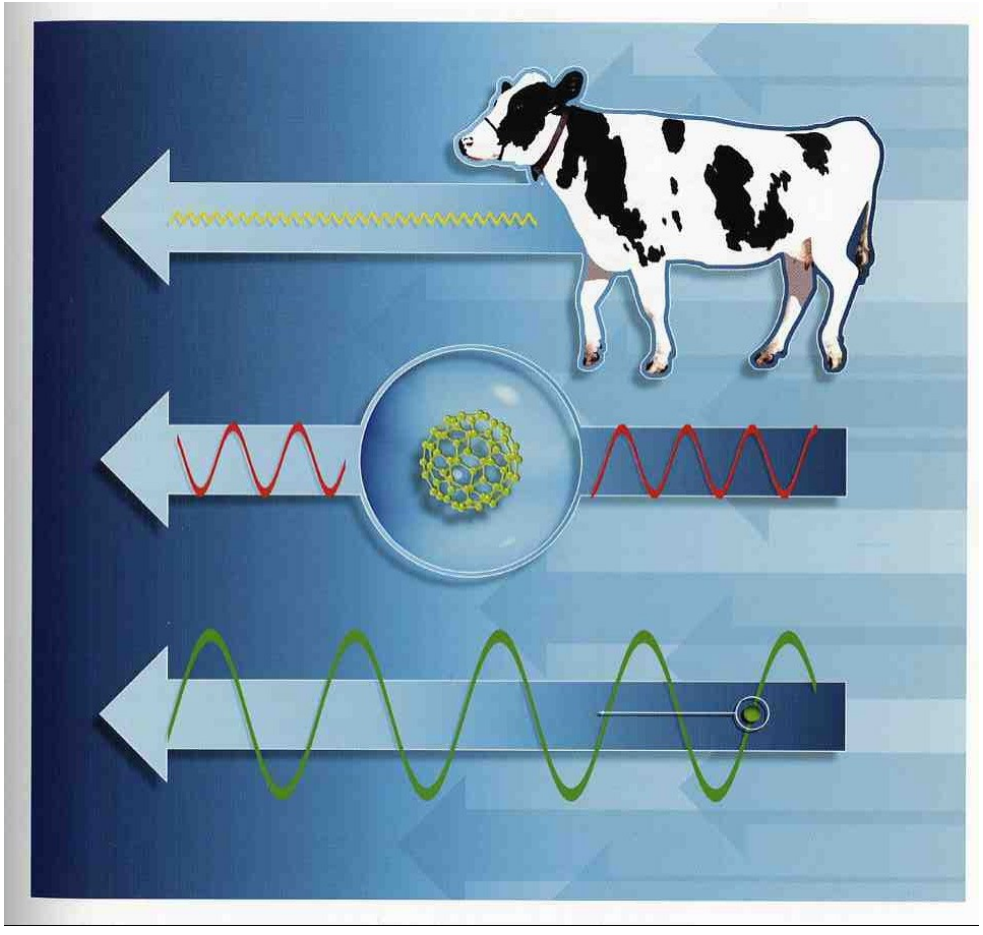
(1 eV= 1.6020×10^{-19} J).

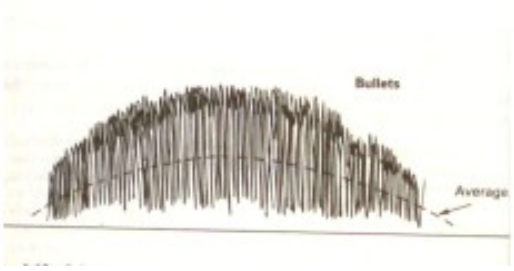
Wave Nature of Particles



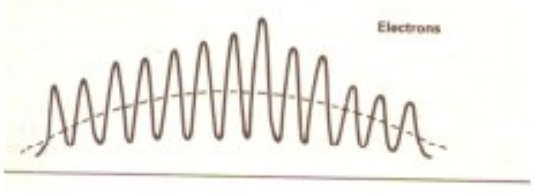
- Louis de Broglie **postulated** that as *light has wave-like and particle-like properties*, *matter must also be both particle-like and wave-like.*
- A particle, of mass m, travelling at velocity v, has linear momentum p = mv.
- By analogy with photons, the associated wavelength of the particle (λ) is given by:

de Broglie's equation $$\lambda = \frac{h}{p} = \frac{h}{mv}$$





- Bullets are far more massive than the electrons. One can observe them as long as one likes but it would not make any difference to them.
- The interference wiggles in the case of bullets are so crowded that it is physically impossible to resolve them, one sees an average behaviour



The wave properties of matter are only apparent for very small masses of matter.

Calculations of Wavelength

Compare the wavelength for an electron (mass = 9.11×10^{-31} kg) traveling at a speed of 1.0×10^{7} m/s with that for a ball (mass = 0.10 kg) traveling at 35 m/s.

Solution We use the equation $\lambda = h/mv$, where

$$h = 6.626 \times 10^{-34}\ \text{J s} \qquad \text{or} \qquad 6.626 \times 10^{-34}\ \text{kg m}^2/\text{s}$$

since

$$1\ \text{J} = 1\ \text{kg m}^2/\text{s}^2$$

For the electron,

$$\lambda_e = \frac{6.626 \times 10^{-34}\ \dfrac{\text{kg m}^2}{\text{s}}}{(9.11 \times 10^{-31}\ \text{kg})(1.0 \times 10^{7}\ \text{m/s})} = 7.3 \times 10^{-11}\ \text{m}$$

For the ball,

$$\lambda_b = \frac{6.626 \times 10^{-34}\ \dfrac{\text{kg m}^2}{\text{s}}}{(0.10\ \text{kg})(35\ \text{m/s})} = 1.9 \times 10^{-34}\ \text{m}$$

Questions

Q1. What will be the wavelength of a ball of mass 0.1 kg moving with a velocity of 10 m s^{-1} ?

Q2. The mass of an electron is 9.1×10^{-31} kg. If its K.E. is 3.0×10^{-25} J, calculate its wavelength.

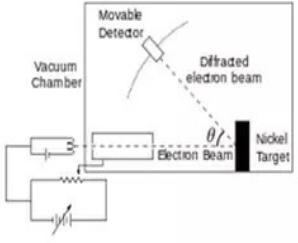
Q3. Calculate the mass of a photon with wavelength 3.6 Å.

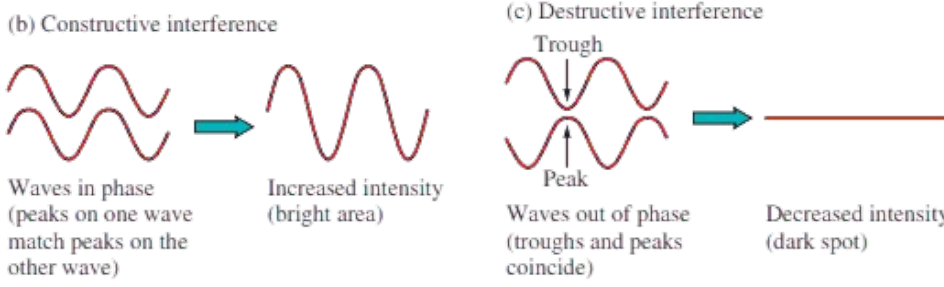
Wave Nature of Particles: Validation

- Davisson and Germer showed that a beam of electrons could be diffracted from the surface of a nickel crystal.
- Scattered light can interfere constructively (the peaks and troughs of the beams are in phase) to produce a bright area or destructively (the peaks and troughs are out of phase) to produce a dark spot.
- ***A diffraction pattern can be explained only in terms of waves.***

DAVISSON-GERMER EXPERIMENT

- The wave nature of electrons was first experimentally verified by C.J.Davisson and L.H.Germer.
- Electrons emitted by the filament are accelerated to desired velocity by applying suitable voltage.
- The electrons are scattered in all directions from the nickel crystal.
- Intensity of scattered electron beam is measured for different values of θ; and later with varying voltages.





Diffraction occurs most efficiently when the spacing between the scattering points is about the same as the wavelength.

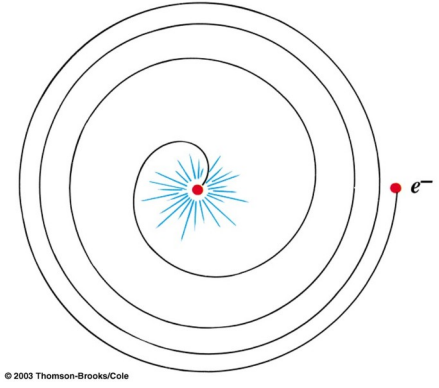
NB: Other "particles" (e.g. neutrons, protons, He atoms) can also be diffracted by crystals.

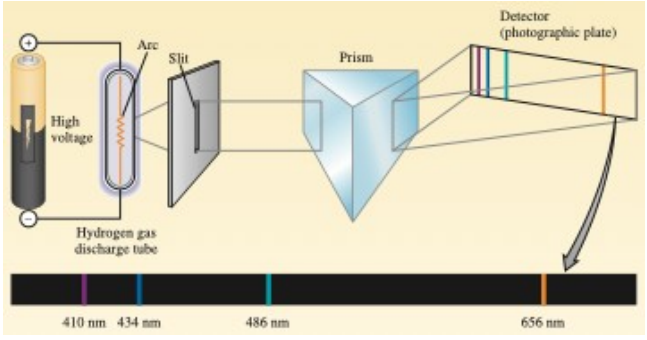
Wave-Particle Duality

- Now we have come full circle.
- Electromagnetic radiation, which at the turn of the twentieth century was thought to be a pure waveform, was found to exhibit particulate properties.
- Conversely, electrons, which were thought to be particles, were found to have a wavelength associated with them.
- *matter and energy are not distinct.*
- Energy is really a form of matter, and all matter shows the same types of properties.

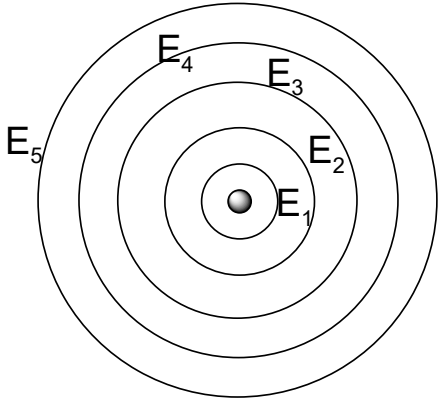
Putting it All together: The Bohr Atom







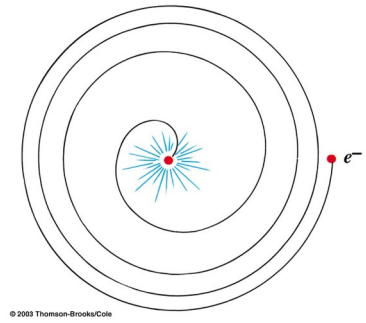
- Bohr proposed a model that included the idea that the *electron in a hydrogen atom moves around the nucleus only in certain allowed circular orbits*.
- Bohr assumed that the hydrogen electron could ***exist only in stationary, non-radiating orbits***.



Putting it All together: The Bohr Atom

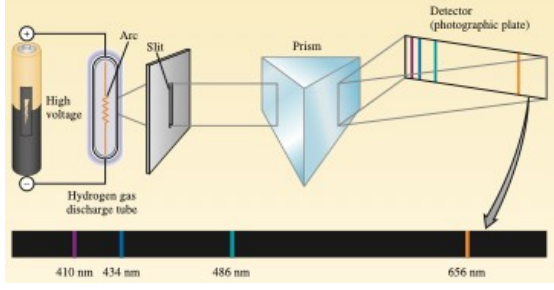
Solving Paradox I: The unstable orbiting electron of Rutherford's atom.

- Make the orbiting electron stable by assuming that the *electron's orbit and energy is quantized to certain values and for these values the orbiting electron does not radiate*.
- The electron is stable in these orbits.
- Thus, the *orbits and energies of electrons are quantized*.



Solving Paradox II: The line spectra of emitting or absorbing atoms.

- Since only certain energies are allowed for orbiting electrons, *only jumps between orbits can be observed*.
- These jumps correspond to *discrete frequencies and wavelengths*.
- Thus, line spectra as expected because of the quantized energies of the orbits.

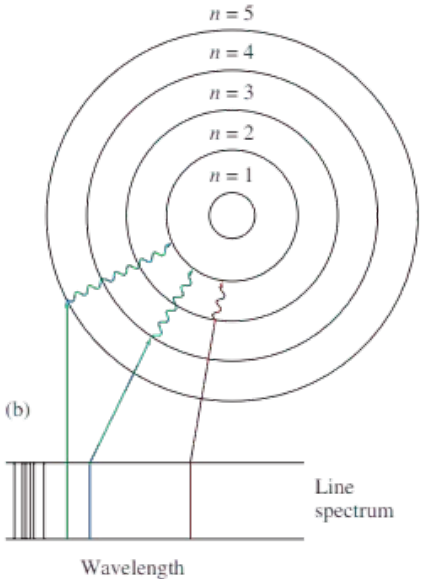


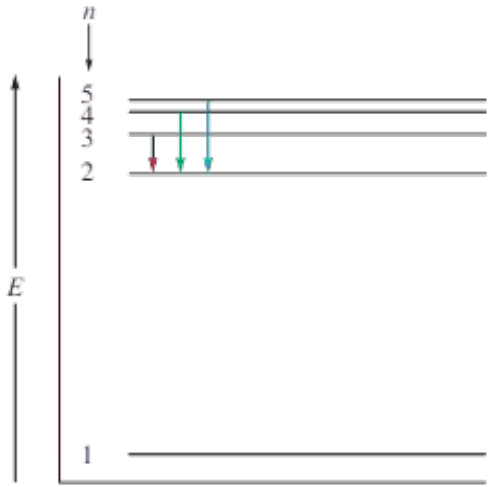
The Bohr Atom

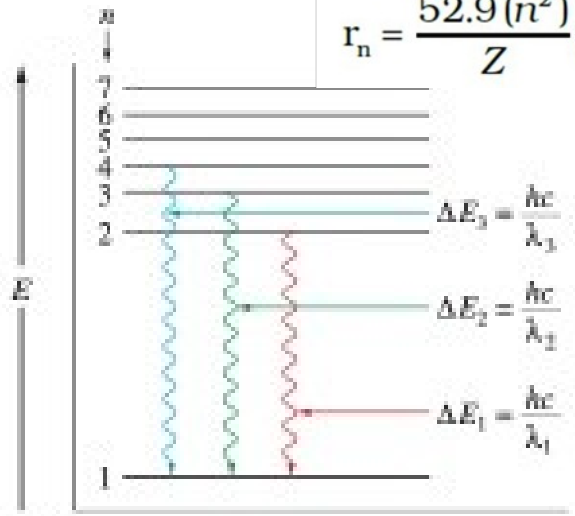
Although we will not discuss its origins here, the expression for the *energy levels available to the electron in the hydrogen atom is*

$$E = -2.178 \times 10^{-18}\,\text{J}\left(\frac{Z^2}{n^2}\right) \quad (12.1)$$

$$r_n = \frac{52.9\,(n^2)}{Z}\,\text{pm}$$





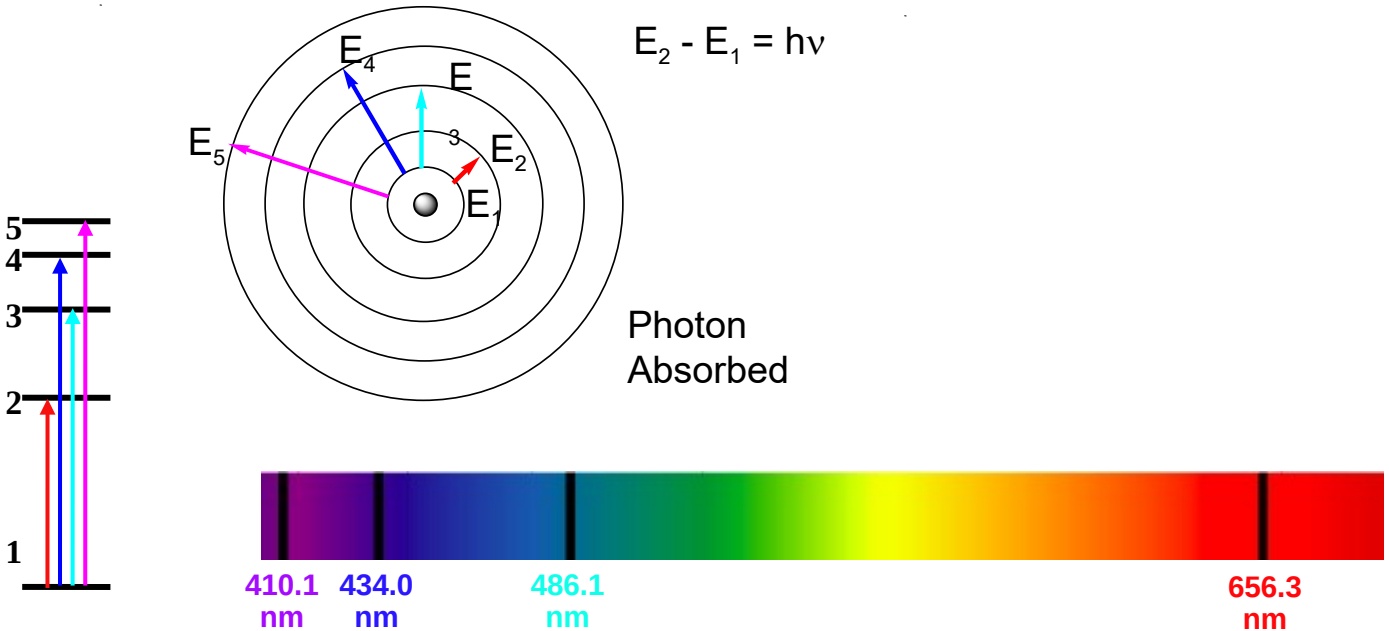


- Light is emitted when an electron jumps from a higher orbit to a lower orbit and absorbed when it jumps from a lower to higher orbit.
- The energy and frequency of light emitted or absorbed is given by the difference between the two orbit energies, e.g.,
- E(photon) = E_2 - E_1 (Energy difference) = h$\boldsymbol{\nu}$

$$\Delta E = h\nu = \frac{hc}{\lambda}$$

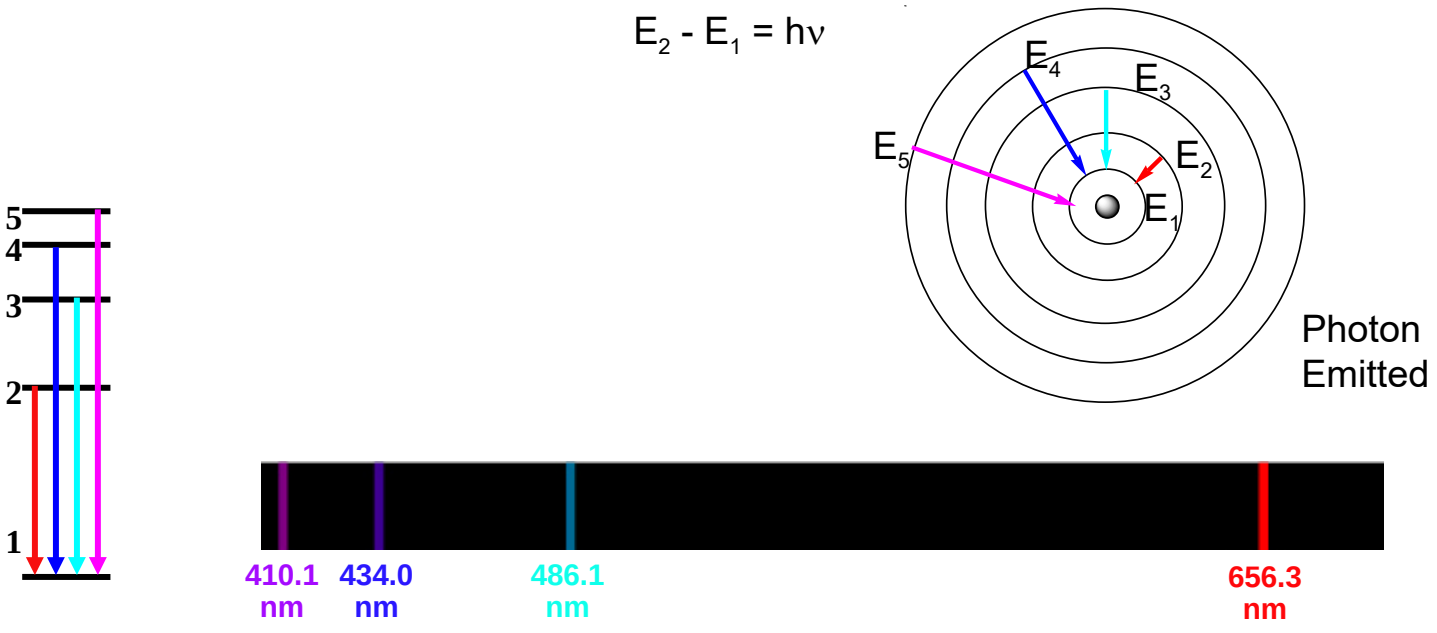
Change in energy; Frequency of light emitted; Wavelength of light emitted

The Bohr Atom



Bohr atom: Light absorption occurs when an electron absorbs a photon and makes a transition for a lower energy orbital to a higher energy orbital. ***Absorption spectra appear as sharp lines.***

The Bohr Atom



Bohr atom: Light emission occurs when an electron makes a transition from a higher energy orbital to a lower energy orbital and a **photon is emitted.** ***Emission spectra appear as sharp lines.***

The Bohr Atom

Limitations of Bohr Model

- At first, Bohr's model appeared to be very promising.
- The energy levels calculated by Bohr closely agreed with the values obtained from the hydrogen emission spectrum.
- However, when Bohr's model was applied to atoms other than hydrogen, it did not work at all
- The model only works for hydrogen (and other one electron ions) – ignores e-e repulsion.
- Does not explain fine structure of spectral lines.
- Note: The Bohr model (assuming circular electron orbits) is fundamentally incorrect.

Questions

Q1. Calculate the energy associated with the first orbit of He^{+} . What is the radius of this orbit?

Q2. How much energy is required to ionise a H atom if the electron occupies n = 5 orbit?

Q3. (i) The energy associated with the first orbit in the hydrogen atom is -2.18×10^{-18} J $atom^{-1}$. What is the energy associated with the fifth orbit?

(ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

Heisenberg's Uncertainty Principle

- Bohr orbit: The electron in the orbital is moving around the nucleus in a circular orbit.
- We can predict the motion of the electron, similar to the motions of particles in the macroscopic world.
- For example, when two billiard balls with known velocities collide, we can predict their motions after the collision.
- We have already learned that an electron does not behave much like a billiard ball.
- Werner Heisenberg, discovered a very important principle in 1927—the **Heisenberg uncertainty principle.**

Heisenberg's Uncertainty Principle

- The **Heisenberg uncertainty principle.**
- *There is a fundamental limitation to just how precisely we can know both the position and the momentum of a particle at a given time.*

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

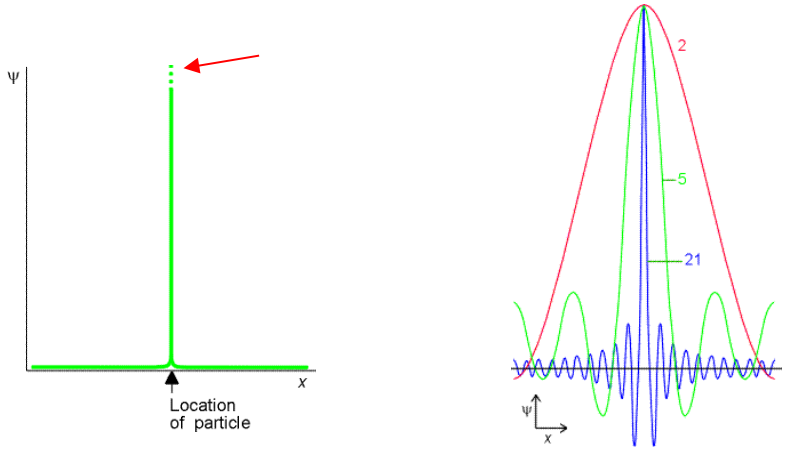
where Δx is the uncertainty in a particle's position, Δp is the uncertainty in a particle's momentum, and ħ is Planck's constant divided by 2π. Thus the minimum uncertainty in the product Δx Δp is $h/4\pi$.

$\hbar = h/2p$

- If we know the position (x) *exactly*, we know nothing about momentum (p_x).
- If we know the momentum (p_x) *exactly*, we know nothing about position (x).

Making Sense of the Uncertainty Principle

- To localize a particle in space (i.e. to specify the particle's position accurately, small Δx) many waves of different wavelengths (λ) must be superimposed $\Rightarrow$ large Δp_x ($p = h/\lambda$).



- The Uncertainty Principle imposes a fundamental (not experimental) limitation on how precisely we can know (or determine) various observables.

The hydrogen atom has a radius on the order of 0.05 nm. Assuming that we know the position of an electron to an accuracy of 1% of the hydrogen radius, calculate the uncertainty in the velocity of the electron using the Heisenberg uncertainty principle. Then compare this value with the uncertainty in the velocity of a ball of mass 0.2 kg and radius 0.05 m known to an accuracy of 1% of its radius.

For the electron the uncertainty in position (Δx) is 1% of 0.05 nm, or

$$\Delta x = (0.01)(0.05\ \text{nm}) = 5. \times 10^{-4}\ \text{nm}$$

Converting to meters gives

$$5 \times 10^{-4}\ \text{nm} \times \frac{10^{-9}\ \text{m}}{1\ \text{nm}} = 5 \times 10^{-13}\ \text{m}$$

The values of the constants are

$$m = \text{mass of the electron} = 9.11 \times 10^{-31}\ \text{kg}$$

$$h = 6.626 \times 10^{-34}\ \text{J s} = 6.626 \times 10^{-34}\ \frac{\text{kg m}^2}{\text{s}}$$

$$\pi = 3.14$$

We can now solve for the uncertainty in momentum:

$$\Delta p = \frac{\hbar}{2 \cdot \Delta x} = \frac{h}{4\pi \cdot \Delta x} = \frac{6.626 \times 10^{-34}\ \frac{\text{kg m}^2}{\text{s}}}{4(3.14)(5 \times 10^{-13}\ \text{m})}$$

$$= 1.05 \times 10^{-22}\ \text{kg m/s} \quad \text{(keeping extra significant figures)}$$

Recalling that $p = mv$ and assuming that the electron mass is constant (ignoring any relativistic corrections), we have

$$\Delta p = \Delta(mv) = m\Delta v$$

and the uncertainty in velocity is

$$\Delta v = \frac{\Delta p}{m} = \frac{1.05 \times 10^{-22}\ \text{kg m/s}}{9.11 \times 10^{-31}\ \text{kg}} = 1.15 \times 10^{8}\ \text{m/s} = 1 \times 10^{8}\ \text{m/s}$$

For the ball, the uncertainty in position (Δx) is 1% of 0.05 m, or 5×10^{-4} m.
Thus the minimum uncertainty in velocity is

$$\Delta v = \frac{\Delta p}{m} = \frac{h}{\Delta x \cdot m \cdot 4\pi} = \frac{6.626 \times 10^{-34}\ \frac{\text{kg m}^2}{\text{s}}}{(5 \times 10^{-4}\ \text{m})(0.2\ \text{kg})(4)(3.14)}$$

$$= 5 \times 10^{-31}\ \text{m/s}$$

Thus the uncertainty principle is negligible in the world of macroscopic objects but is very important for objects with small masses, such as the electron.

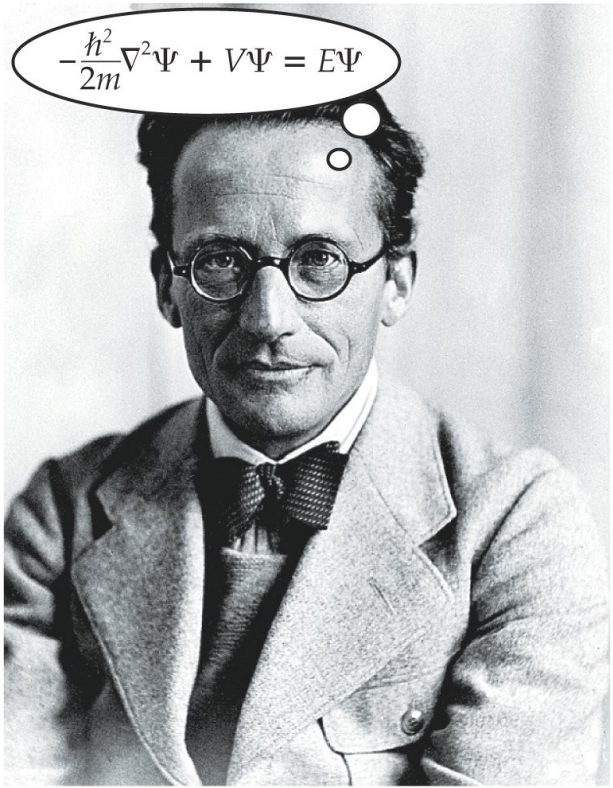
- This limitation is so small for large particles such as baseballs or billiard balls that it is unnoticed.
- However, for a small particle such as the electron, the limitation becomes quite important.
- Applied to the electron, the uncertainty principle implies that we cannot know the exact path of the electron as it moves around the nucleus.
- *It is therefore not appropriate to assume that the electron is moving around the nucleus in a well-defined orbit as in the Bohr model.*

Questions

Q1. A microscope using suitable photons is employed to locate an electron in an atom within a distance of 0.1 Å. What is the uncertainty involved in the measurement of its velocity?

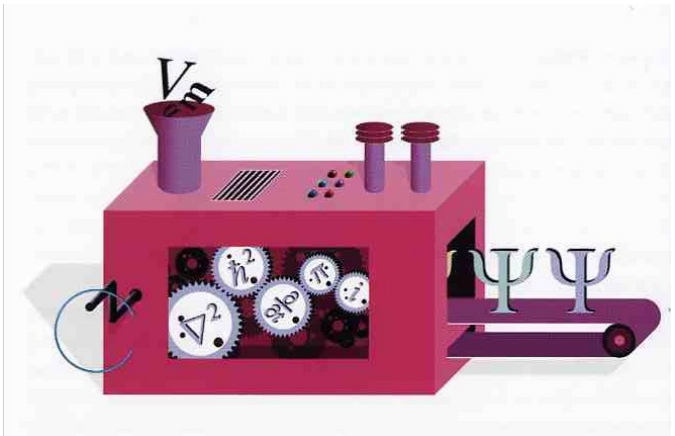
Q2. A golf ball has a mass of 40g, and a speed of 45 m/s. If the speed can be measured within accuracy of 2%, calculate the uncertainty in the position.

Quantum Mechanics



Schrödinger: If electrons are waves, their position and motion in space must obey a wave equation.

Solutions of wave equations yield wavefunctions, Ψ, which contain the information required to describe ALL of the properties of the wave.



The Schrödinger Equation (1926)

- The central equation in Quantum Mechanics.
- Observable = total energy of system.

Schrödinger Equation $\hat{\mathbf{H}}\psi = E\psi$ Hamiltonian Operator $\hat{\mathbf{H}}$

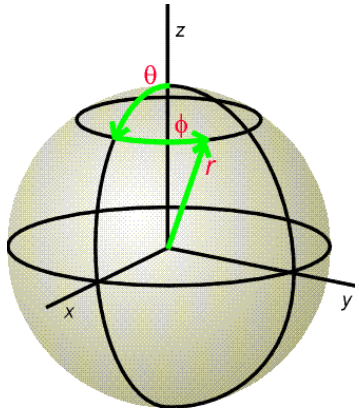
E Total Energy

where $\hat{\mathbf{H}} = \hat{\mathbf{T}} + \hat{\mathbf{V}}$ and $E = T + V$.

- SE can be set up for any physical system.
- The form of $\hat{\mathbf{H}}$ depends on the system.
- Solve SE $\Rightarrow$ ψ and corresponding E.

Wavefunctions

- A particle trajectory is a classical concept.
- In Quantum Mechanics, a "particle" (e.g. an electron) does not follow a definite trajectory $\{\mathbf{r}(t),\mathbf{p}(t)\}$, but rather it is best described as being distributed through space like a wave.
- Wavefunction (ψ) – a wave representing the spatial distribution of a "particle".
  - e.g. electrons in an atom are described by a wavefunction centred on the nucleus.
- ψ is a function of the coordinates defining the position of the classical particle:
  - 1-D $\psi(x)$
  - 3-D $\psi(x,y,z) = \psi(\mathbf{r}) = \psi(r,\theta,\phi)$ (e.g. atoms)
- ψ may be time dependent – e.g. $\psi(x,y,z,t)$



Wavefunctions

The Importance of ψ

- ψ completely defines the system (e.g. electron in an atom or molecule).
- If ψ is known, we can determine any observable property (e.g. energy, vibrational frequencies, …) of the system.
- QM provides the tools to determine ψ computationally, to interpret ψ and to use ψ to determine properties of the system.

Wavefunctions

Interpretation of the Wavefunction

- In QM, a "particle" is distributed in space like a wave.
- We cannot define a position for the particle.
- Instead we define a probability of finding the particle at any point in space.

The Born Interpretation (1926)

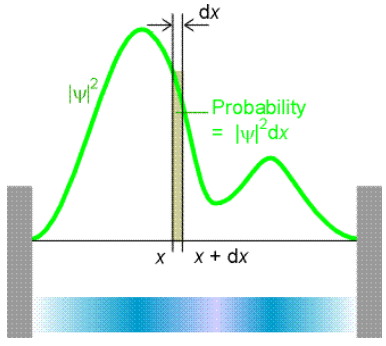
"The square of the wavefunction at any point in space is proportional to the probability of finding the particle at that point."

- Note: *the wavefunction (ψ) itself has no physical meaning*.

Example: 1-D System

- If the wavefunction at point x is $\psi(x)$, the probability of finding the particle in the infinitesimally small region (dx) between x and x+dx is:

$$P(x) \propto \underbrace{|\psi(x)|^2}_{\text{probability density}} dx$$



- $|\psi(x)|$ – the magnitude of ψ at point x.

Why write $|\psi|^2$ instead of ψ^2 ?

- Because ψ may be imaginary or complex $\Rightarrow$ ψ^2 would be negative or complex.
- BUT: probability must be real and positive ($0 \le P \le 1$).
- For the general case, where ψ is complex ($\psi = a + ib$) then:

$|\psi|^2 = \psi^*\psi$ where ψ^* is the complex conjugate of ψ.

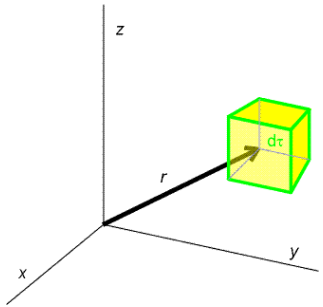
($\psi^* = a - ib$) (NB $i = \sqrt{-1}$)

Example: 3-D System

- If the wavefunction at $\mathbf{r} = (x,y,z)$ is $\psi(\mathbf{r})$, the probability of finding the particle in the infinitesimal volume element $d\tau$ (= dxdydz) is:

$$P(\mathbf{r}) \propto |\psi(\mathbf{r})|^2 \, d\tau$$

- If $\psi(\mathbf{r})$ is the wavefunction describing the spatial distribution of an electron in an atom or molecule, then:



$|\psi(\mathbf{r})|^2 = \rho(\mathbf{r})$ – the electron density at point $\mathbf{r}$

Questions

Q1. The wavefunction of an electron in the lowest energy state of a hydrogen atom is proportional to $e^{-\frac{r}{a_0}}$, with a_0 a constant and r the distance from the nucleus. Calculate the relative probabilities of finding the electron inside a region of volume $\delta V = 1.0\ \text{pm}^3$, which is small even on the scale of the atom, located at (a) the nucleus, (b) a distance a_0 from the nucleus.

Normalization of the Wavefunction

- As mentioned above, probability: $P(\mathbf{r}) \propto |\psi(\mathbf{r})|^2 \, d\tau$
- What is the proportionality constant?
- If ψ is such that the sum of $|\psi(\mathbf{r})|^2$ at all points in space = 1, then:

$$P(x) = |\psi(x)|^2 \, dx \qquad \text{1-D}$$

$$P(\mathbf{r}) = |\psi(\mathbf{r})|^2 \, d\tau \qquad \text{3-D}$$

- As summing over an infinite number of infinitesimal steps = integration, this means:

$$P_{total}(1D) = \int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1$$

$$P_{total}(3D) = \int_{-\infty}^{\infty} |\psi(\mathbf{r})|^2 \, d\tau = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\psi(x, y, z)|^2 \, dx dy dz = 1$$

- i.e. the probability that the particle is somewhere in space = 1.
- In this case, ψ is said to be a normalized wavefunction.

How to Normalize the Wavefunction

- If ψ is not normalized, then:

$$\int |\psi(\mathbf{r})|^2 \, d\tau = A \neq 1$$

- A corresponding normalized wavefunction (ψ_{Norm}) can be defined:

$$\psi_{Norm}(\mathbf{r}) = \frac{1}{\sqrt{A}} \psi(\mathbf{r})$$

such that:

$$\int |\psi_{Norm}(\mathbf{r})|^2 \, d\tau = 1$$

- The factor ($1/\sqrt{A}$) is known as the normalization constant (sometimes represented by N).

Quantization of the Wavefunction

The Born interpretation of ψ places restrictions on the form of the wavefunction:

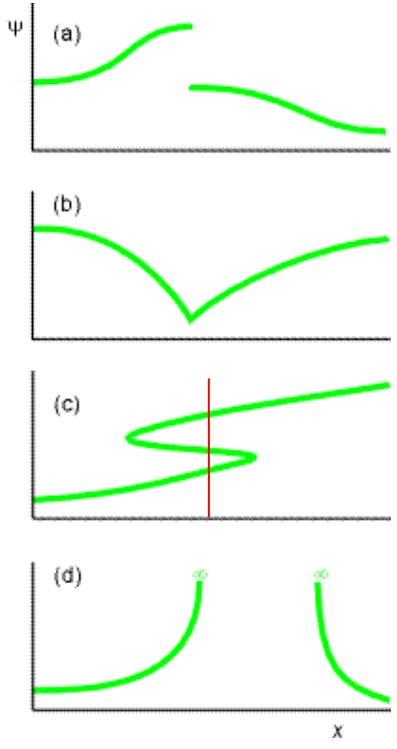
(a) ψ must be continuous (no breaks);

(b) The gradient of ψ (dψ/dx) must be continuous (no kinks);

(c) ψ must have a single value at any point in space;

(d) ψ must be finite everywhere;

(e) ψ cannot be zero everywhere.



- Other restrictions (boundary conditions) depend on the exact system.
- These restrictions on ψ mean that only certain wavefunctions and only certain energies of the system are allowed.

Quantization of ψ ⇒ Quantization of E

Question

Q1. The wavefunction of an electron in the lowest energy state of a hydrogen atom is proportional to $e^{-\frac{r}{a_0}}$, with a_0 a constant and r the distance from the nucleus. Normalize the wavefunction.

Question

Q1. The wavefunction for the electron in its lowest energy state in the ion He^+ is proportional to $e^{-\frac{2r}{a_0}}$. Normalize the wavefunction.

Hamiltonian Operator

- Wavefunction (ψ) contains all the information we need to know about any particular system.
- How do we determine ψ and use it to deduce properties of the system?

Operators and Observables

- If ψ is the wavefunction representing a system, we can write:

$$\hat{\mathbf{Q}}\psi = Q\psi$$

where Q – "observable" property of system (e.g. energy, momentum, dipole moment …)

$\hat{\mathbf{Q}}$ – operator corresponding to observable Q.

- This is an eigenvalue equation and can be rewritten as:

$$\hat{\mathbf{Q}}(\psi) = Q \times \psi$$

operator **Q** acting on function ψ (eigenfunction)

function ψ multiplied by a number Q (eigenvalue)

(Note: ψ can't be cancelled).

Examples: $d/dx\ (e^{ax}) = a\ e^{ax}$

$$d^2/dx^2\ (\sin ax) = -a^2 \sin ax$$

Questions

Q1. Show that e^{ax} is an eigenfunction of the operator $\frac{d}{dx}$, and find the corresponding eigenvalue. Show that e^{ax^2} is not an eigenfunction of $\frac{d}{dx}$.

Q2. Is the function $\cos ax$ an eigenfunction of (a) $\frac{d}{dx}$, (b) $\frac{d^2}{dx^2}$?

To find ψ and calculate the properties (observables) of a system:

1. Construct relevant operator $\hat{\mathbf{Q}}$
2. Set up equation $\hat{\mathbf{Q}}\psi = Q\psi$
3. Solve equation $\Rightarrow$ allowed values of ψ and Q.

Quantum Mechanical Position and Momentum Operators

1. Operator for position in the x-direction is just multiplication by x

$$\hat{\mathbf{x}}\psi = x\psi$$

2. Operator for linear momentum in the x-direction: $\hat{\mathbf{p}}_x = \left(\frac{\hbar}{i}\right)\frac{d}{dx}$

$$\hat{\mathbf{p}}_x\psi = p_x\psi \qquad \frac{\hbar}{i}\cdot\frac{d\psi}{dx} = p_x\psi$$

(solve first order differential equation $\Rightarrow$ ψ , p_x).

Constructing Kinetic and Potential Energy QM Operators

1. Write down classical expression in terms of position and momentum.
2. Introduce QM operators for position and momentum.

Examples

1. Kinetic Energy Operator in 1-D $\hat{T}_x$

CM $T_x = \frac{p_x^2}{2m}$ $\Rightarrow$ QM $\hat{T}_x = \frac{\hat{p}_x^2}{2m} = -\frac{\hbar^2}{2m}\left(\frac{d^2}{dx^2}\right)$

2. KE Operator in 3-D $\hat{T}$

CM $T = \frac{p^2}{2m} = \frac{p_x^2 + p_y^2 + p_z^2}{2m}$

QM $\hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) = -\frac{\hbar^2}{2m}\nabla^2$

"del-squared"

partial derivatives operate on $\psi(x,y,z)$

3. Potential Energy Operator $\hat{V}$ (a function of position)

$\therefore$ PE operator corresponds to multiplication by V(x), V(x,y,z) etc.

Examples

1. Particle Moving in 1-D $\psi(x)$

$$\hat{\mathbf{H}}\psi = \hat{\mathbf{T}}\psi + \hat{\mathbf{V}}\psi = E\psi \longrightarrow -\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}\right) + V(x)\psi = E\psi$$

- The form of $V(x)$ depends on the physical situation:
  - Free particle $V(x) = 0$ for all x.
  - Harmonic oscillator $V(x) = \frac{1}{2}kx^2$

2. Particle Moving in 3-D $\psi(x,y,z)$

- SE $\Rightarrow$

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}\right) + V(x,y,z)\psi = E\psi$$

or

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(x,y,z)\psi = E\psi$$

Note: The SE is a second order differential equation

Question

Q1. The operator for the angular momentum of a particle travelling in a circle in the xy-plane is $\hat{l}_z = \frac{\hbar}{i}\frac{d}{d\phi}$, where φ is its angular position. What is the angular momentum of a particle described by the wavefunction $e^{-2i\phi}$?

Particle in a Box model

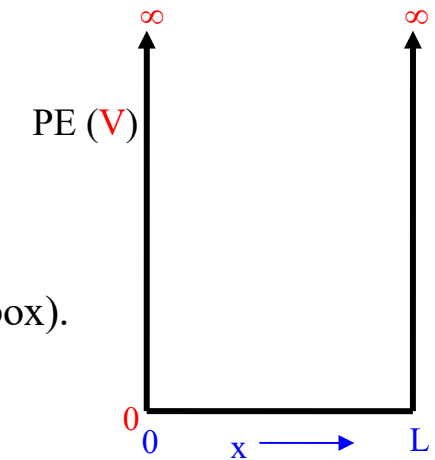
System

- Particle of mass m in 1-D box of length L.
- Position $x = 0 \rightarrow L$.
- Particle cannot escape from box as PE $V(x) = \infty$ for $x = 0, L$ (walls).
- PE inside box: $V(x) = 0$ for $0 < x < L$.

1-D Schrödinger Eqn.

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}\right) = E\psi$$

(V = 0 inside box).



$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}\right)=E\psi$$

- This is a second order differential equation – with general solutions of the form:

$$\psi = A\sin kx + B\cos kx$$

$$\left(\frac{\partial^2\psi}{\partial x^2}\right)=-k^2\left(A\sin kx+B\cos kx\right)=-k^2\psi$$

- SE $\Rightarrow$

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2}\right)=\left(-\frac{\hbar^2}{2m}\right)\cdot\left(-k^2\psi\right)=E\psi$$

$\Rightarrow$ $\boxed{E=\dfrac{k^2\hbar^2}{2m}}$ (i.e. E depends on k).

Restrictions on ψ

- In principle Schrödinger Eqn. has an infinite number of solutions.
- So far we have general solutions:
  - any value of $\{A, B, k\} \rightarrow$ any value of $\{\psi, E\}$.
- BUT – due to the Born interpretation of ψ, only certain values of ψ are physically acceptable:
  - outside box ($x<0$, $x>L$) $V = \infty \Rightarrow$ impossible for particle to be outside the box

    $\therefore \ |\psi|^2 = 0 \quad \Rightarrow \quad \psi = 0$ outside box.
  - ψ must be a continuous function

    $\Rightarrow$ Boundary Conditions $\psi = 0$ at $x = 0$

    $\psi = 0$ at $x = L$.

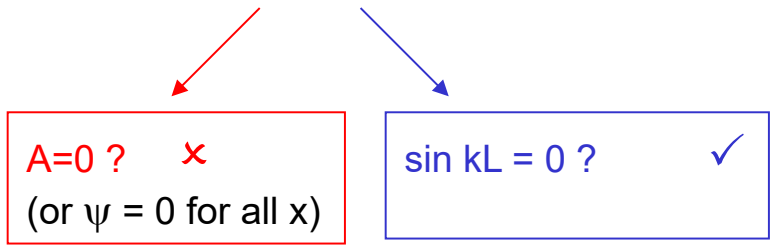
Effect of Boundary Conditions

1. $x = 0$ $\quad \psi = A \underbrace{\sin kx}_{0} + B \underbrace{\cos kx}_{1} = B$

$\psi = 0 \quad \therefore B = 0$

$\Rightarrow \quad \psi = A \sin kx \quad$ for all x

2. $x = L \quad \psi = A \sin kL = 0$



$\sin kL = 0 \Rightarrow \quad kL = n\pi \quad n = 1, 2, 3, \ldots$

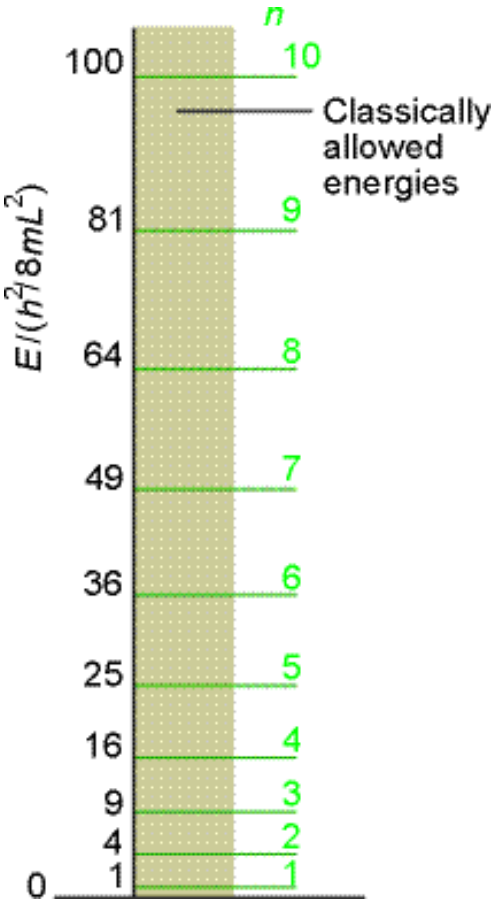
Allowed Wavefunctions and Energies

- k is restricted to a discrete set of values: $k = n\pi/L$
- Allowed wavefunctions: $\psi_n = A \sin(n\pi x/L)$
- Normalization: $A = \sqrt{(2/L)} \Rightarrow$ $$\psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$
- Allowed energies: $$E_n = \frac{k^2\hbar^2}{2m} = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

$$\Rightarrow \quad E_n = \frac{n^2 h^2}{8mL^2}$$

Quantum Numbers

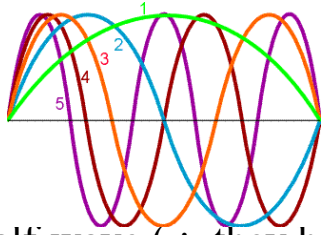
- There is a discrete energy state (E_n), corresponding to a discrete wavefunction (ψ_n), for each integer value of n.
- Quantization – occurs due to boundary conditions and requirement for ψ to be physically reasonable (Born interpretation).
- n is a Quantum Number – labels each allowed state (ψ_n) of the system and determines its energy (E_n).
- Knowing n, we can calculate ψ_n and E_n.



Properties of the Wavefunction

- Wavefunctions are standing waves: $$\psi_n = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

- The first 5 normalized wavefunctions for the particle in the 1-D box:



- Successive functions possess one more half-wave ($\therefore$ they have a shorter wavelength).

- Nodes in the wavefunction – points at which $\psi_n = 0$ (excluding the ends which are constrained to be zero).

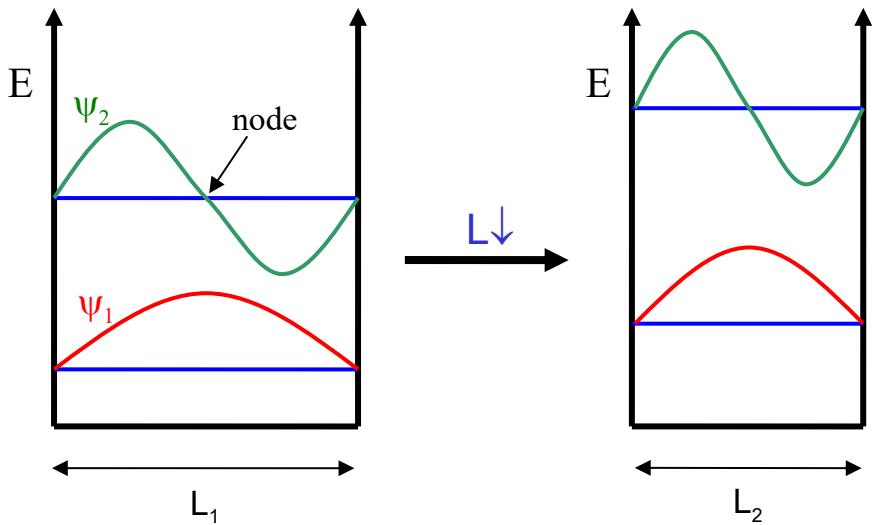
- Number of nodes = (n-1) $\quad \psi_1 \rightarrow 0;\ \psi_2 \rightarrow 1;\ \psi_3 \rightarrow 2 \ldots$

Energies $E_n = \frac{n^2 h^2}{8mL^2}$

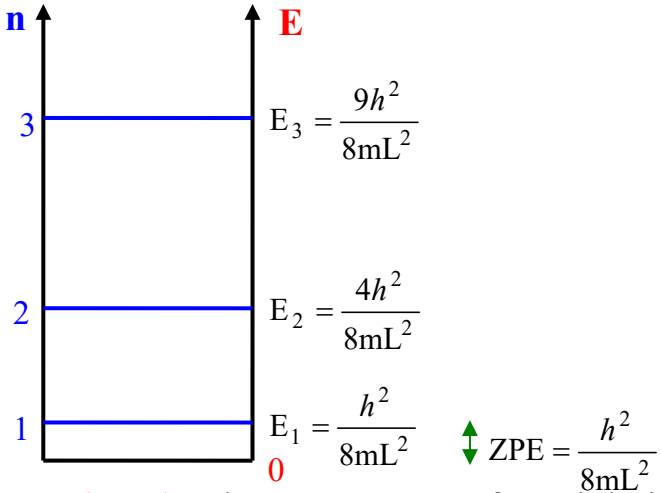
- $E_n \propto n^2/L^2$ $\Rightarrow$ $E_n\uparrow$ as $n\uparrow$ (more nodes in ψ_n)
  $E_n\uparrow$ as $L\downarrow$ (shorter box)

  $n\uparrow$ (or $L\downarrow$) $\Rightarrow$ curvature of $\psi_n\uparrow$

  $\Rightarrow$ KE$\uparrow$ $\Rightarrow$ $E_n\uparrow$



- $E_n \propto n^2 \Rightarrow$ energy levels get further apart as n↑



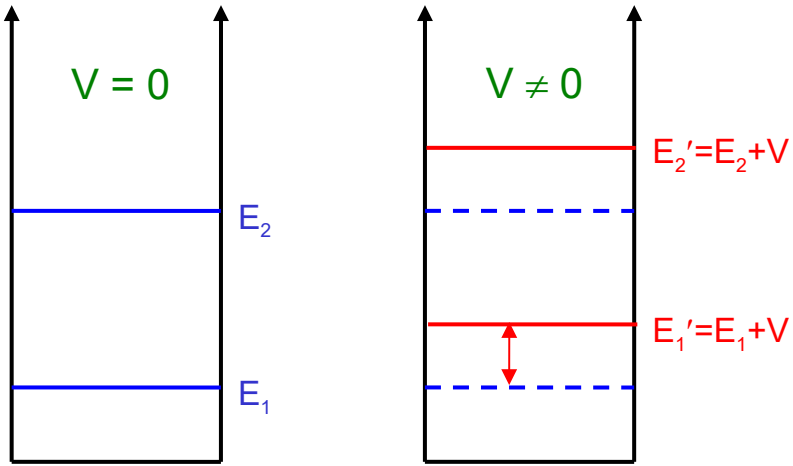
- Zero-Point Energy (ZPE) – lowest energy of particle in box:

$$ZPE = E_{min} = E_1 = \frac{h^2}{8mL^2}$$

- CM $E_{min} = 0$
- QM $E = 0$ corresponds to $\psi = 0$ everywhere (forbidden).

- If $V(x) = V \neq 0$, everywhere in box, all energies are shifted by V.

$$E_n = \frac{n^2 h^2}{8mL^2} + V$$

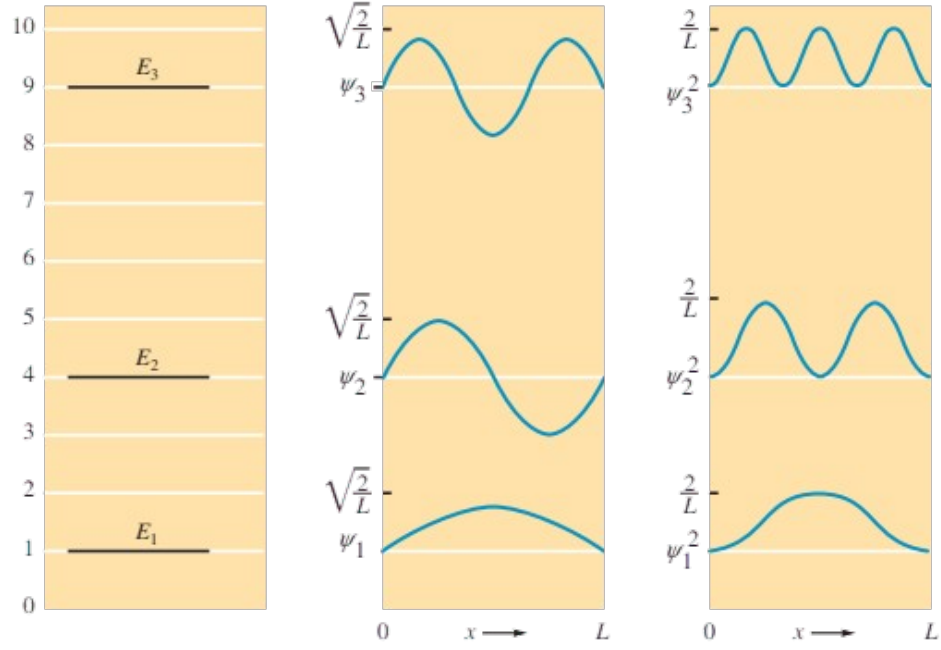


Density Distribution of the Particle in the 1-D Box

- The probability of finding the particle between x and x+dx (in the state represented by ψ_n) is:

  $P_n(x) = |\psi_n(x)|^2\,dx = (\psi_n(x))^2\,dx$ (ψ_n is real)

$$\Rightarrow \quad P_n(x) = \frac{2}{L}\sin^2\left(\frac{n\pi x}{L}\right)dx$$

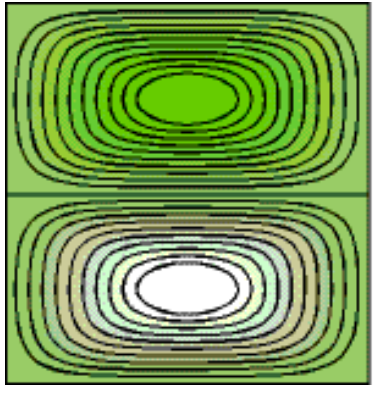


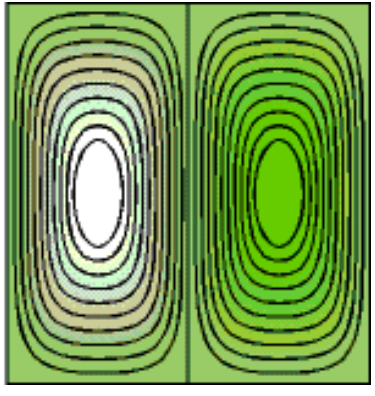
- Note: probability is not uniform
  - $\psi_n^2 = 0$ at walls (x = 0, L) for all ψ_n.
  - $\psi_n^2 = 0$ at nodes (where $\psi_n = 0$).

Particle in a 2-D Square or 3-D Cubic Box

- Similar to 1-D case, but $\psi \rightarrow \psi(x,y)$ or $\psi(x,y,z)$.
- Solutions are now defined by 2 or 3 quantum numbers

  e.g. $[\psi_{n,m}, E_{n,m}]$; $[\psi_{n,m,l}, E_{n,m,l}]$.
- Wavefunctions can be represented as contour plots in 2-D

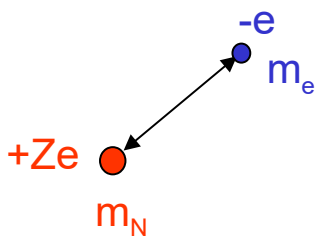




Question

Hydrogen Atom

- The series of atoms/ions H, He^+, Li^{2+}, Be^{3+} …
  all have: 1 electron (charge –e)
  nucleus (charge +Ze)



- To determine the electronic wavefunction ($\psi = \psi_e$) and allowed electronic energy levels (E), we must set up and solve the Schrödinger Equation for a single electron in an atom.

$$\hat{H} = \hat{T} + \hat{V}$$

$$= \left(-\frac{h^2}{8\pi^2 m_e}\nabla_e^2 - \frac{h^2}{8\pi^2 m_n}\nabla_n^2 \right) - \frac{Ze^2}{(4\pi\varepsilon_0)r}$$

$$\left[-\frac{h^2}{8\pi^2 m_e}\left(\frac{\partial^2}{\partial x_e^2}+\frac{\partial^2}{\partial y_e^2}+\frac{\partial^2}{\partial z_e^2}\right) - \frac{h^2}{8\pi^2 m_n}\left(\frac{\partial^2}{\partial x_n^2}+\frac{\partial^2}{\partial y_n^2}+\frac{\partial^2}{\partial z_n^2}\right) - \frac{Ze^2}{(4\pi\varepsilon_0)r} \right] \Psi_{total} = E_{total}\,\Psi_{total}$$

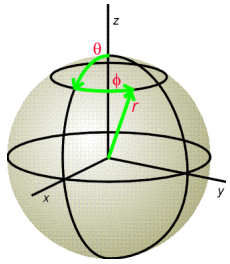
$$-\frac{h^2}{8\pi^2(m_e+m_n)}\left(\frac{\partial^2}{\partial x_c^2}+\frac{\partial^2}{\partial y_c^2}+\frac{\partial^2}{\partial z_c^2}\right)\psi_M = E_{trans}\psi_M$$

Free movement of the center of mass of the atom in space

$$\left[-\frac{h^2}{8\pi^2\mu_e}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right) - \frac{Ze^2}{(4\pi\varepsilon_0)r} \right]\psi_e = E\,\psi_e$$

Relative motion of the electron with respect to the nucleus within the atom

$$E_{trans} + E = E_{total}$$



- We must solve the 2nd order differential equation:

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi + V(r)\psi = E\psi$$

- Because of the spherical symmetry of the atom, it is convenient to describe the position of the electron in spherical polar coordinates (r,θ,ϕ), rather than Cartesians (x,y,z).

- Due to this spherical symmetry, the wavefunction can be separated into a product of radial and angular components:

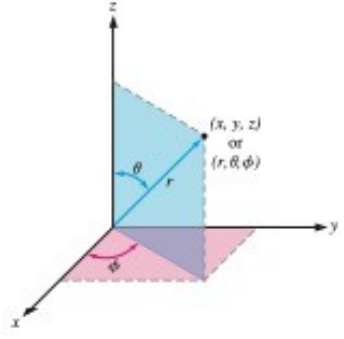
$$\psi(r,\theta,\phi) = R(r).Y(\theta,\phi)$$

radial wavefunction

angular wavefunction (spherical harmonic)

- Imposing boundary conditions $\Rightarrow$ 3 quantum numbers (n, ℓ, m_ℓ)

$$\Rightarrow \qquad \psi_{n,\ell,m\ell}(r,\theta,\phi) = R_{n,\ell}(r).Y_{\ell,m\ell}(\theta,\phi)$$



$$\psi(r,\theta,\phi) = R(r).Y(\theta,\phi)$$

TABLE 1.2 Hydrogen Wavefunctions (Atomic Orbitals), $\psi = RY$

(a) Radial wavefunctions, $R_{nl}(r)$

n	l	$R_{nl}(r)$
1	0	$2\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/a_0}$
2	0	$\frac{1}{2\sqrt{2}}\left(\frac{Z}{a_0}\right)^{3/2}\left(2 - \frac{Zr}{a_0}\right)e^{-Zr/2a_0}$
	1	$\frac{1}{2\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right)e^{-Zr/2a_0}$
3	0	$\frac{1}{9\sqrt{3}}\left(\frac{Z}{a_0}\right)^{3/2}\left(3 - \frac{2Zr}{a_0} + \frac{2Z^2r^2}{9a_0^2}\right)e^{-Zr/3a_0}$
	1	$\frac{2}{27\sqrt{6}}\left(\frac{Z}{a_0}\right)^{3/2}\left(2 - \frac{Zr}{3a_0}\right)e^{-Zr/3a_0}$
	2	$\frac{4}{81\sqrt{30}}\left(\frac{Z}{a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right)^2 e^{-Zr/3a_0}$

(b) Angular wavefunctions, $Y_{lm_l}(\theta,\phi)$

l	"m_l"*	$Y_{lm_l}(\theta,\phi)$
0	0	$\left(\frac{1}{4\pi}\right)^{1/2}$
1	x	$\left(\frac{3}{4\pi}\right)^{1/2} \sin\theta\cos\phi$
	y	$\left(\frac{3}{4\pi}\right)^{1/2} \sin\theta\sin\phi$
	z	$\left(\frac{3}{4\pi}\right)^{1/2} \cos\theta$
2	xy	$\left(\frac{15}{16\pi}\right)^{1/2} \sin^2\theta\cos 2\phi$
	yz	$\left(\frac{15}{4\pi}\right)^{1/2} \cos\theta\sin\theta\sin\phi$
	zx	$\left(\frac{15}{4\pi}\right)^{1/2} \cos\theta\sin\theta\cos\phi$
	$x^2 - y^2$	$\left(\frac{15}{16\pi}\right)^{1/2} \sin^2\theta\sin 2\phi$
	z^2	$\left(\frac{5}{16\pi}\right)^{1/2} (3\cos^2\theta - 1)$

Note: In each case, $a_0 = 4\pi\varepsilon_0{}^2/m_e e^2$, or close to 52.9 pm; for hydrogen itself, $Z = 1$.
*In all cases except $m_l = 0$, the orbitals are sums and differences of orbitals with specific values of m_l.

Quantum Numbers

$$\psi_{n,\ell,m\ell}(r,\theta,\phi) = R_{n,\ell}(r).Y_{\ell,m\ell}(\theta,\phi)$$

- ψ depends on 3 quantum numbers (n, ℓ, m_ℓ).

1. Principal Quantum Number (n)

- Positive integer $n = 1, 2, 3 \ldots$
- For H and 1-e ions, the electron energy depends only on n

$$E_n = -\left(\frac{\mu e^4}{8h^2\varepsilon_0^2}\right)\cdot\left(\frac{Z^2}{n^2}\right)$$

2. Orbital Angular Momentum Quantum Number (ℓ)

- For a given value of n, ℓ can take the integer values:

$$\ell = 0, 1, 2 \ldots (n\text{-}1)$$

e.g.

$n = 1$	$\ell = 0$
$n = 2$	$\ell = 0, 1$
$n = 3$	$\ell = 0, 1, 2$

3. Orbital Magnetic Quantum Number (m_ℓ)

- For a given value of ℓ, m_ℓ can take the integer values:

$$m_\ell = 0, \pm 1, \pm 2 \ldots \pm \ell$$

e.g.

$\ell = 0$	$m_\ell = 0$
$\ell = 1$	$m_\ell = 0, \pm 1$

- There are $(2\ell+1)$ allowed values of m_ℓ.

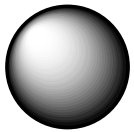
Quantum Numbers and Atomic Orbitals

- Quantum theory postulates that electrons in atoms are fixed in regions of space corresponding to atomic orbitals.
- The probability of finding an electron at a particular point in space ($\mathbf{r}$) is proportional to $|\psi(\mathbf{r})|^2$ (Born Interpretation).

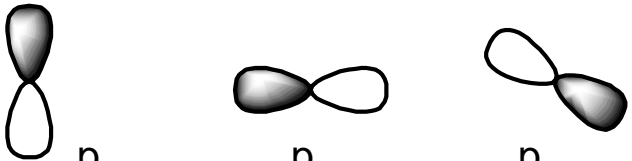
- Orbitals differ from each other in *size, shape and orientation*:
  - the orbital size is defined by the *principle quantum number* n
  - the type of orbital (it's shape) is defined by the angular momentum quantum number ℓ
  - the orientation of the orbital is defined by the orbital magnetic quantum number m_ℓ

Shapes of Atomic Orbitals

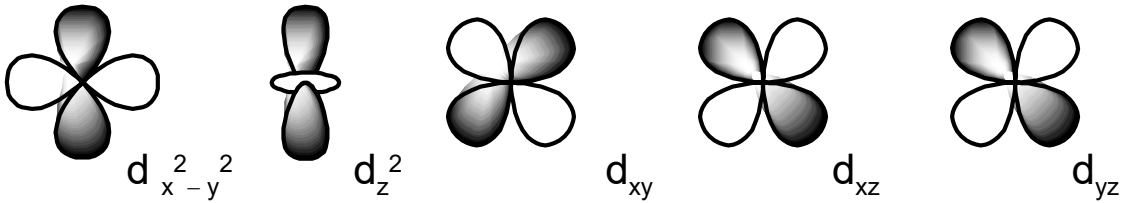
- s orbitals have $l = 0$ and $m_l = 0$



- p orbitals have $l = 1$ and $m_l = 1, 0, -1$



- d orbitals have $l = 2$ and $m_l = 2, 1, 0, -1, -2$

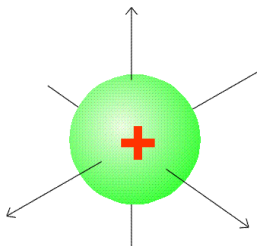


Shape of Atomic Orbitals

Shape – determined by the angular wavefunction $Y_{\ell,m\ell}(\theta,\phi)$

1s orbital (n = 1, ℓ = 0, |J| = 0)

- Normalized wavefunction: $\psi(1s) = \frac{1}{\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}$
- Has no angular dependence (spherically symmetric, depends only on r).
- All s orbitals are spherically symmetrical.



2s orbital

$$\psi(2s) \propto (2-r)e^{-r/2}$$

ns orbital

$$\psi(ns) \propto f(r)e^{-r/n}$$

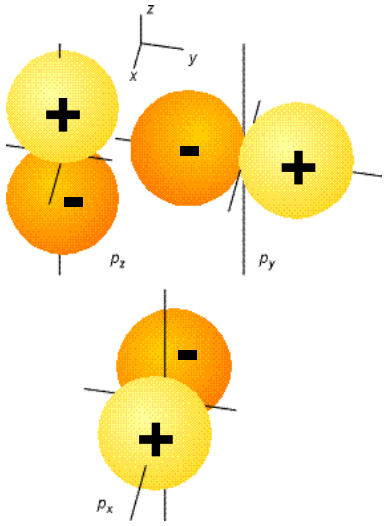
- As n↑, orbitals expand – average radius ⟨r⟩↑

2p orbitals (n = 2, ℓ = 1, |J| = $\sqrt{2}\hbar$)

- There are 3 degenerate orbitals, with 3 different m_ℓ values (0,±1).

- e.g. $m_\ell = 0 \Rightarrow J_z = m_\ell \hbar = 0 \Rightarrow 2p_z$ orbital (pointing along z-axis)

$$\psi(2p_z) = \frac{1}{4\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{5/2} \cdot r\cos\theta \cdot e^{-r/2a_0}$$

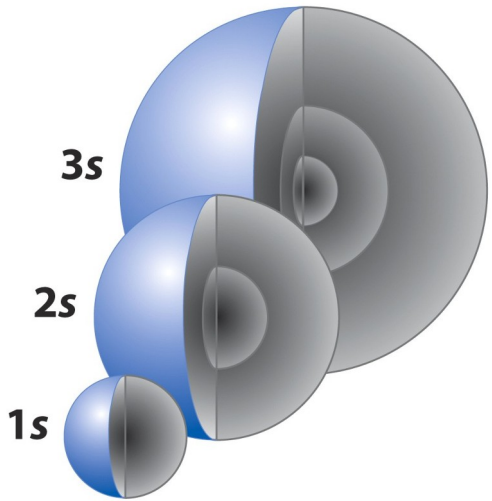


- $m_\ell = \pm 1 \Rightarrow J_z = \pm\hbar \Rightarrow 2p_x, 2p_y$ orbitals

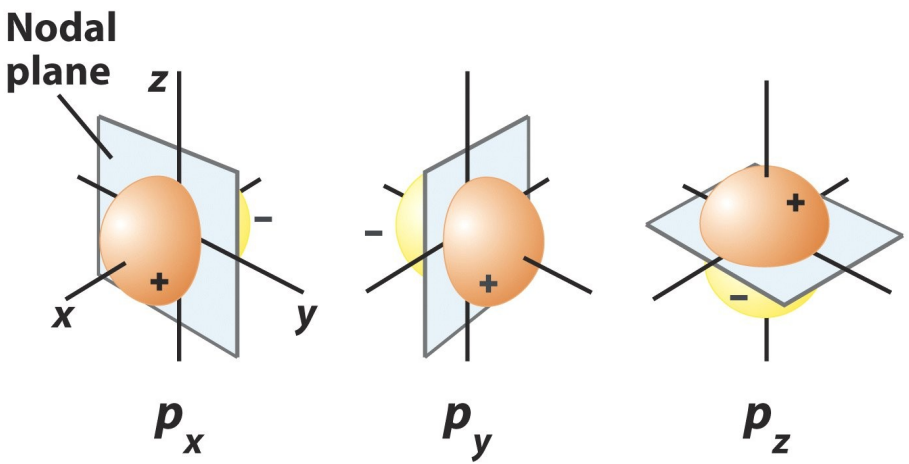
Wavefunctions and orbitals

- An orbital is a wavefunction
- Orbital: defined by the quantum numbers n, l and m_l (which are solutions of the wave equation)
- Orbital is a region of space occupied by an electron
- Orbitals has energies, shapes and orientation in space

s orbitals



p orbitals



Atomic Orbitals

- The wavefunctions ($\psi_{n,\ell,m\ell}$) describing an electron in an atom are known as atomic orbitals.

- Each atomic orbital (1-e wavefunction) is uniquely defined by 3 quantum numbers (n,ℓ,m_ℓ).

- The orbital (ψ) gives the spatial distribution of the electron (via the Born interpretation of $|\psi|^2$).

- Orbitals are often drawn as 3-D surfaces which enclose approx. 90% of the probability of finding the electron.

Electronic Shells, Sub-shells and Orbitals

- All orbitals with the same value of n together constitute an electronic shell.

n	1	2	3	4
Shell	K	L	M	N

- For each n, orbitals with the same value of ℓ together constitute an electronic sub-shell.

- Each sub-shell consists of $(2\ell+1)$ orbitals, each with a different m_ℓ value.

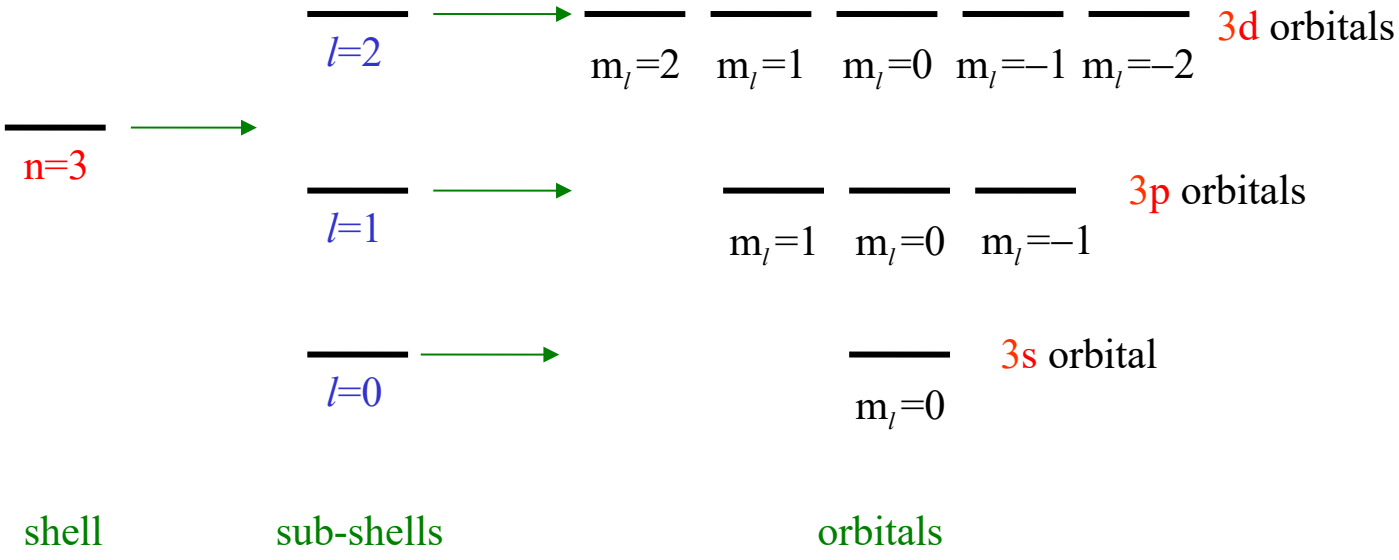
ℓ	0	1	2	3
Sub-shell	s	p	d	f
No. orbitals	1	3	5	7

- Possible combinations of shells and sub-shells.

$\ell \rightarrow$ n	0	1	2	3	…
1	1s				
2	2s	2p			
3	3s	3p	3d		
4	4s	4p	4d	4f	
…					

- Total number of orbitals in shell n is n^2.

Electronic Shells, Sub-shells and Orbitals

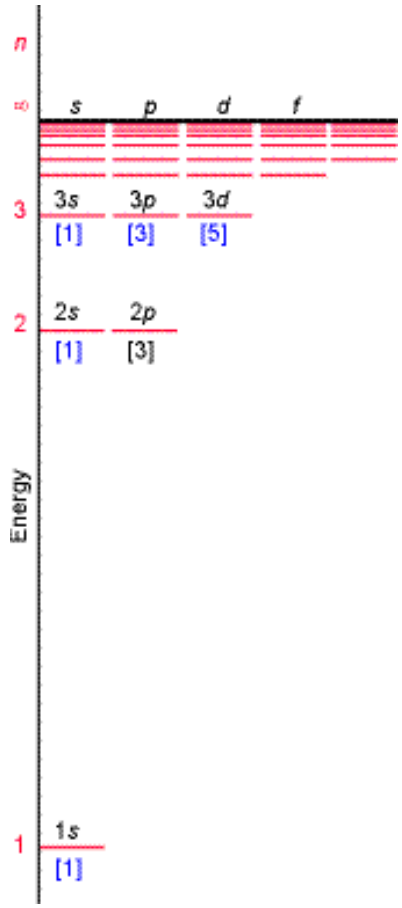


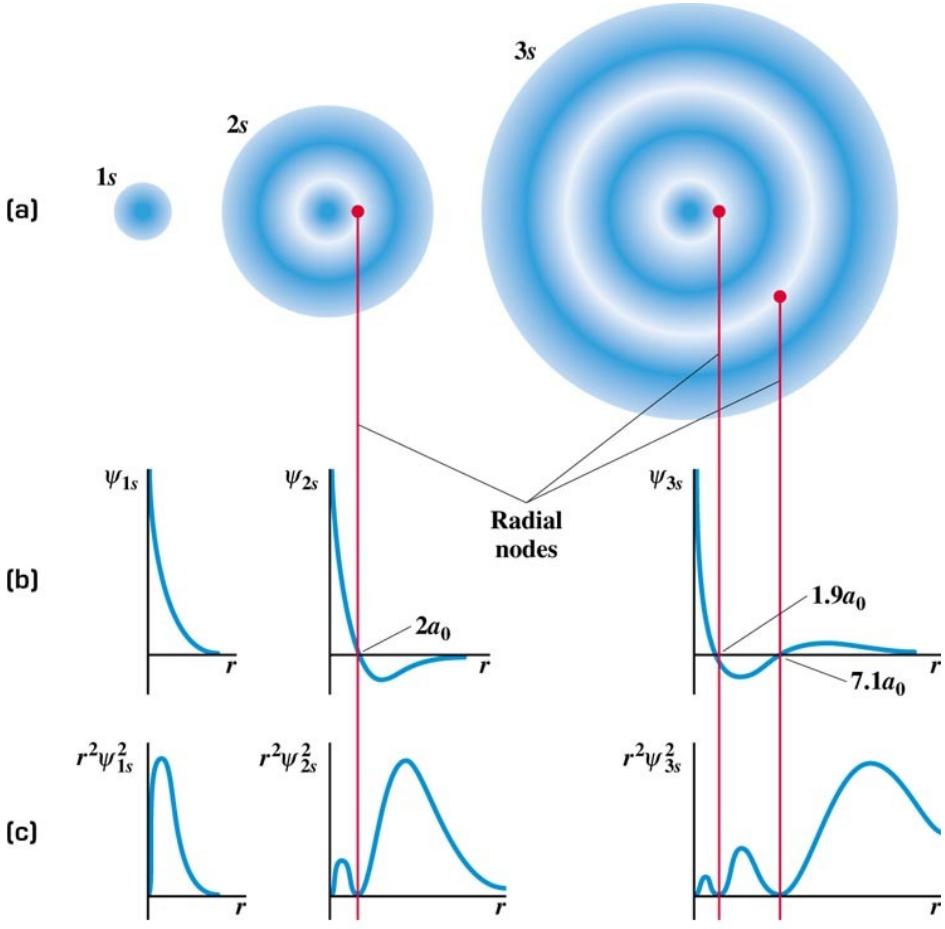
Atomic Orbital Energy Diagram for H

- Orbital energies depend only on the principal quantum number n.

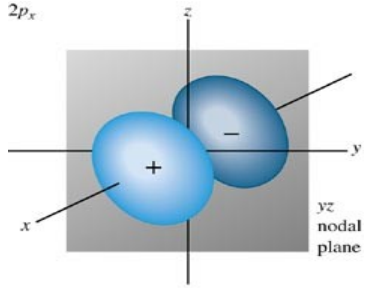
- For a given n value (shell), all sub-shells (ℓ) and orbitals (m_ℓ) have the same energy

  – i.e. they are degenerate.

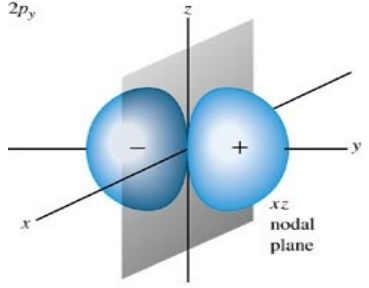




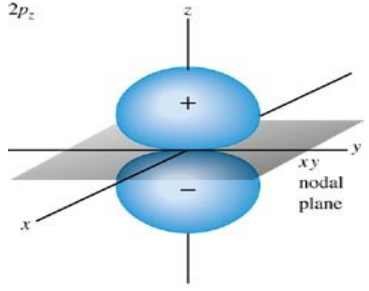
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p_x



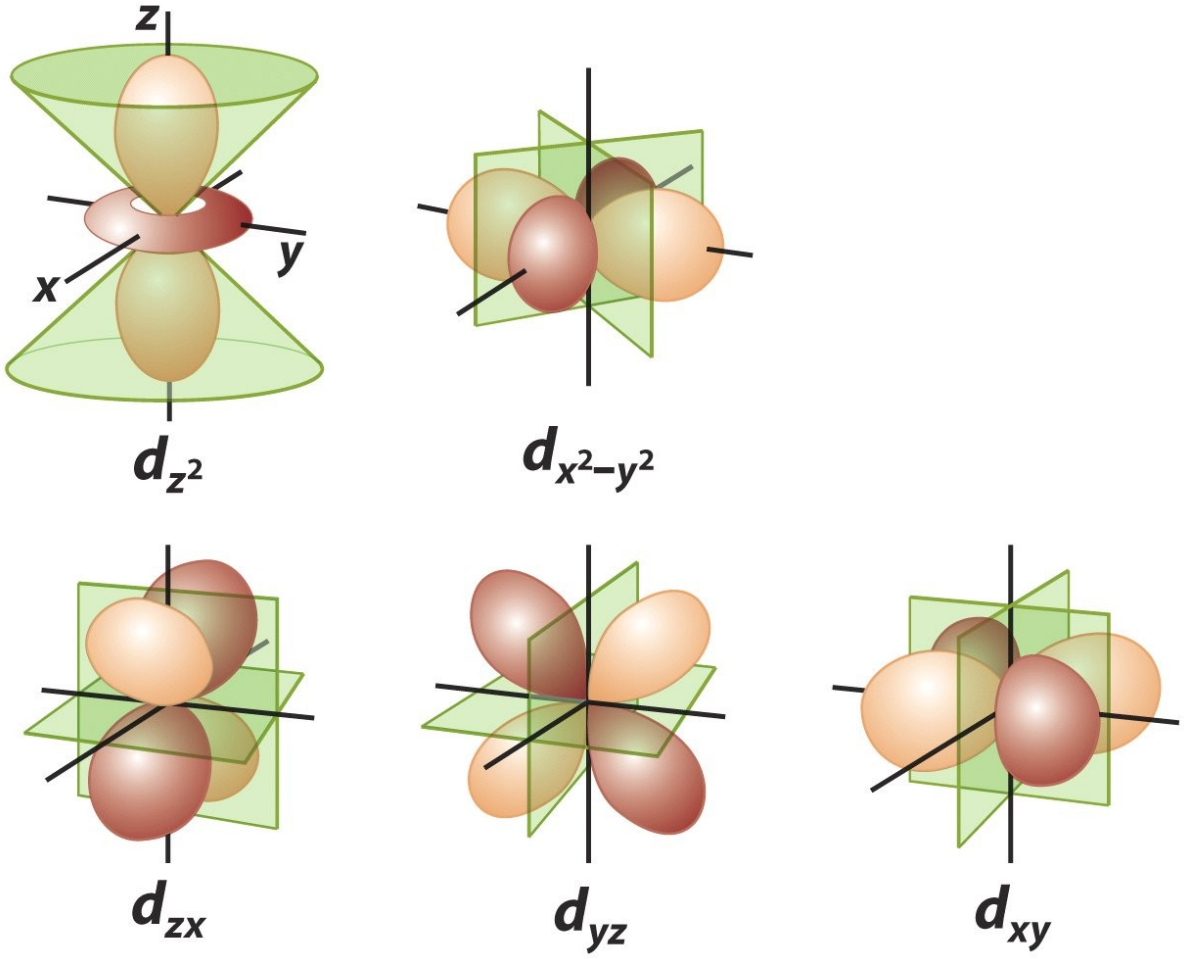
p_y



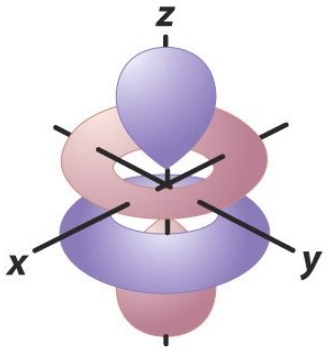
p_z

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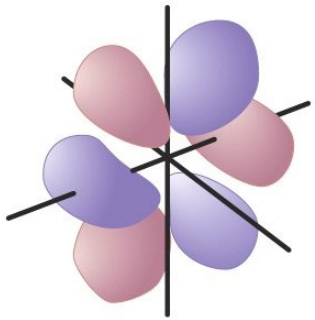
The d orbitals of a one electron atom



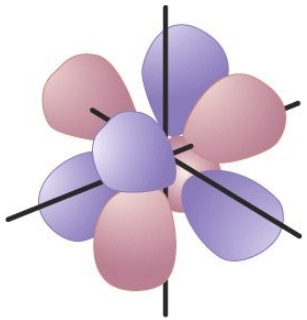
The f orbitals of a one electron atom



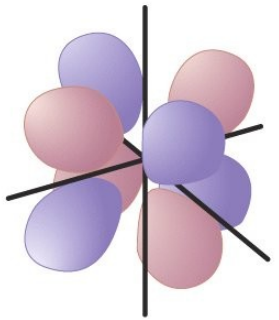
$5z^3 - 3zr^2$



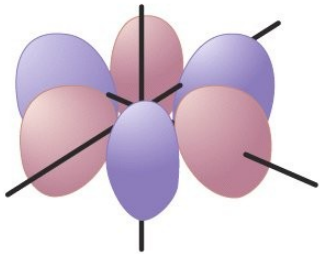
$5xz^2 - xr^2$



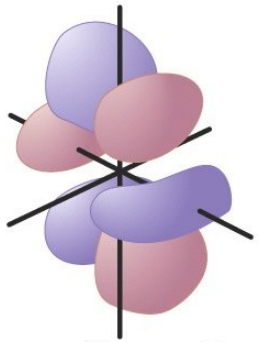
$zx^2 - zy^2$



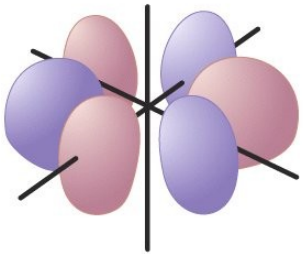
xyz



$y^3 - 3yx^2$



$5yz^2 - yr^2$



$x^3 - 3xy^2$

Quantum Numbers (QN)

Principal QN:

n = 1, 2, 3, 4……

Angular momentum QN:

l = 0, 1, 2, 3…. (n -1)

Rule: *l* = (n - 1)

Magnetic QN:

m_l = …-2, -1, 0, 1, 2, ..

Rule: -*l*….0….+*l*

TABLE 1.3 Quantum Numbers for Electrons in Atoms

Name	Symbol	Values	Specifies	Indicates
principal	n	$1, 2, \ldots$	shell	size
orbital angular momentum*	l	$0, 1, \ldots, n-1$	subshell: $l = 0, 1, 2, 3, 4, \ldots$ $s, p, d, f, g, \ldots$	shape
magnetic	m_l	$l, l-1, \ldots, -l$	orbitals of subshell	orientation
spin magnetic	m_s	$+\frac{1}{2}, -\frac{1}{2}$	spin state	spin direction

*Also called the *azimuthal quantum number.*

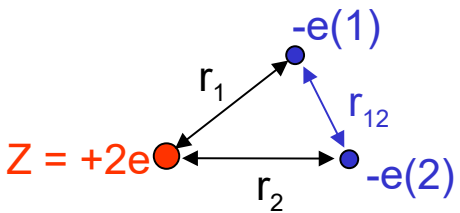
Many-Electron Atoms

- For atoms with 2 or more electrons, the Schrödinger Eqn. must include KE terms for all electrons and PE terms for all e-e and e-n interactions.

- Example: He (2 electrons)

2 terms in $\hat{T}$

3 terms in $\hat{V}$



S.E.

$$-\frac{\hbar^2}{2m_e}\nabla_1^2-\frac{\hbar^2}{2m_e}\nabla_2^2+\left(-\frac{2e^2}{4\pi\varepsilon_0 r_1}-\frac{2e^2}{4\pi\varepsilon_0 r_2}+\frac{e^2}{4\pi\varepsilon_0 r_{12}}\right)\psi=E\psi$$

where

$$\nabla_i^2=\left(\frac{\partial^2\psi}{\partial x_i^2}+\frac{\partial^2\psi}{\partial y_i^2}+\frac{\partial^2\psi}{\partial z_i^2}\right)$$

- For N electrons: $\Psi_N = \Psi(\mathbf{r}_1,\mathbf{r}_2,\mathbf{r}_3 \ldots \mathbf{r}_N)$

- For more than 1 e, S.E. cannot be solved analytically – though good numerical solutions may be obtained from computer calculations.

$\Rightarrow$ Approximations must be made.

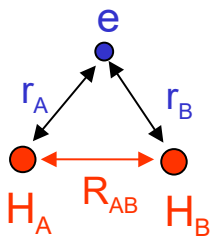
The Orbital Approximation

- The total wavefunction (Ψ_N) for the N-electron atom is approximated by the product of N 1-e orbitals similar to those of hydrogenic ions:

$$\Psi_N = \Psi(\mathbf{r}_1,\mathbf{r}_2,\mathbf{r}_3 \ldots \mathbf{r}_N) = \psi(\mathbf{r}_1)\psi(\mathbf{r}_2)\psi(\mathbf{r}_3)\ldots\psi(\mathbf{r}_N)$$

Bonding in Molecules

Exact solution of the Schrödinger Equation is not possible for any molecule – even the simplest molecule H_2^+.



Full Hamiltonian operator for H_2^+:

KE PE

$$\hat{H} = -\frac{\hbar^2}{2m_p}\left(\nabla_A^2 + \nabla_B^2\right) - \frac{\hbar^2}{2m_e}\nabla_e^2 - \frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{r_A} + \frac{1}{r_B}\right) + \frac{e^2}{4\pi\varepsilon_0 R_{AB}}$$

nuclear K.E. electron K.E. e-n attraction n-n repulsion

Application

- Computational chemistry is a rapidly growing field in chemistry.
  - Computers are getting faster.
  - Algorithms and programs are maturing.

- Some of the almost limitless properties that can be calculated with computational chemistry are:
  - Equilibrium and transition-state structures
  - dipole and quadrupole moments and polarizabilities
  - Vibrational frequencies, IR and Raman Spectra
  - NMR spectra
  - Electronic excitations and UV spectra
  - Reaction rates and cross sections
  - thermochemical data

Limitations

- Schrödinger Equation can only be solved exactly for simple systems.
  - Rigid Rotor, Harmonic Oscillator, Particle in a Box, Hydrogen Atom
- For more complex systems (*i.e.* many electron atoms/molecules) we need to make some simplifying assumptions/approximations and solve it numerically.
- However, it is still possible to get very accurate results (and also get very crummy results).
  - In general, the "cost" of the calculation increases with the accuracy of the calculation and the size of the system.

Tools

- There exist a large number of software packages capable of performing electronic structure calculations.
  - MOLPRO, GAMESS, COLUMBUS, NWCHEM, MOLFDIR, ACESII, GAUSSIAN, ...
  - The different programs have various advantages and capabilities.
- We also have available to us Gaussview which is a GUI that interfaces with Gaussian for aiding in building molecules and viewing output.

Thank you!